

A. 4000 bytes B. 8000 bytes C. 10000 bytes D. 12000 bytes 1

gateit-2004 computer-networks sliding-window normal 2

Answer key 

2.30.15 Sliding Window: GATE IT 2006 | Question: 64

Suppose that it takes 1 unit of time to transmit a packet (of fixed size) on a communication link. The link layer uses a window flow control protocol with a window size of N packets. Each packet causes an ack or a nak to be generated by the receiver, and ack/nak transmission times are negligible. Further, the round trip time on the link is equal to N units. Consider time $i > N$. If only acks have been received till time i (no naks), then the goodput evaluated at the transmitter at time i (in packets per unit time) is 3

- A. $1 - \frac{N}{i}$ B. $\frac{i}{(N + i)}$ 4
C. 1 D. $1 - e^{(-\frac{i}{N})}$

gateit-2006 computer-networks sliding-window normal 5

Answer key  6

2.30.16 Sliding Window: GATE IT 2008 | Question: 64

A 1 Mbps satellite link connects two ground stations. The altitude of the satellite is 36,504 km and speed of the signal is 3×10^8 m/s. What should be the packet size for a channel utilization of 25% for a satellite link using go-back-127 sliding window protocol? Assume that the acknowledgment packets are negligible in size and that there are no errors during communication. 7

- A. 120 bytes B. 60 bytes C. 240 bytes D. 90 bytes 8

gateit-2008 computer-networks sliding-window normal 9

Answer key 

2.31 Slotted Aloha (1)

2.31.1 Slotted Aloha: GATE CSE 2015 | Set 1 | Question: 29

Consider a LAN with four nodes S_1, S_2, S_3 , and S_4 . Time is divided into fixed-size slots, and a node can begin its transmission only at the beginning of a slot. A collision is said to have occurred if more than one node transmits in the same slot. The probabilities of generation of a frame in a time slot by S_1, S_2, S_3 , and S_4 are 0.1, 0.2, 0.3 and 0.4 respectively. The probability of sending a frame in the first slot without any collision by any of these four stations is 10

gatecse-2015-set1 computer-networks normal numerical-answers slotted-aloha 11

Answer key 

2.32 Sockets (4)

2.32.1 Sockets: GATE CSE 2008 | Question: 17

Which of the following system calls results in the sending of SYN packets? 

- A. socket B. bind C. listen D. connect

gatecse-2008 normal computer-networks sockets

Answer key 

2.32.2 Sockets: GATE CSE 2008 | Question: 59

A client process P needs to make a TCP connection to a server process S. Consider the following situation: the server process S executes a `socket()`, a `bind()` and a `listen()` system call in that order, following which it is preempted. Subsequently, the client process P executes a `socket()` system call followed by `connect()` system call to connect to the server process S. The server process has not executed any `accept()` system call. Which one 

of the following events could take place? 1

- A. `connect()` system call returns successfully 2
- B. `connect()` system call blocks
- C. `connect()` system call returns an error
- D. `connect()` system call results in a core dump

gatecse-2008 computer-networks sockets normal 3

Answer key

2.32.3 Sockets: GATE CSE 2014 | Set 2 | Question: 24

Which of the following socket API functions converts an unconnected active TCP socket into a passive socket? 4

- A. `connect`
- B. `bind`
- C. `listen`
- D. `accept` 5

gatecse-2014-set2 computer-networks sockets easy 6

Answer key

2.32.4 Sockets: GATE CSE 2015 | Set 2 | Question: 20

Identify the correct order in which a server process must invoke the function calls `accept`, `bind`, `listen`, and `recv` according to UNIX socket API. 7

- A. `listen`, `accept`, `bind`, `recv`
- B. `bind`, `listen`, `accept`, `recv` 8
- C. `bind`, `accept`, `listen`, `recv`
- D. `accept`, `listen`, `bind`, `recv`

gatecse-2015-set2 computer-networks sockets easy 9

Answer key 10

2.33 Stop and Wait (6)

Practice Test: Test 1 (13Q) 11

2.33.1 Stop and Wait: GATE CSE 2015 | Set 1 | Question: 53

Suppose that the stop-and-wait protocol is used on a link with a bit rate of 64 kilobits per second and 20 milliseconds propagation delay. Assume that the transmission time for the acknowledgment and the processing time at nodes are negligible. Then the minimum frame size in bytes to achieve a link utilization of at least 50 % is _____. 12

gatecse-2015-set1 computer-networks stop-and-wait normal numerical-answers 13

Answer key

2.33.2 Stop and Wait: GATE CSE 2016 | Set 1 | Question: 55

A sender uses the Stop-and-Wait ARQ protocol for reliable transmission of frames. Frames are of size 1000 bytes and the transmission rate at the sender is 80 Kbps (1Kbps = 1000 bits/second). Size of an acknowledgment is 100 bytes and the transmission rate at the receiver is 8 Kbps. The one-way propagation delay is 100 milliseconds. 14

Assuming no frame is lost, the sender throughput is _____ bytes/ second. 15

gatecse-2016-set1 computer-networks stop-and-wait normal numerical-answers 16

Answer key

2.33.3 Stop and Wait: GATE CSE 2017 | Set 1 | Question: 45

The values of parameters for the Stop-and-Wait ARQ protocol are as given below:

- Bit rate of the transmission channel = 1 Mbps.
- Propagation delay from sender to receiver = 0.75 ms.

- Time to process a frame = 0.25 ms.
- Number of bytes in the information frame = 1980.
- Number of bytes in the acknowledge frame = 20.
- Number of overhead bytes in the information frame = 20.

1

Assume there are no transmission errors. Then, the transmission efficiency (expressed in percentage) of the Stop-and-Wait ARQ protocol for the above parameters is _____ (correct to 2 decimal places).

gatecse-2017-set1 computer-networks stop-and-wait numerical-answers normal 3

Answer key

2.33.4 Stop and Wait: GATE CSE 2023 | Question: 7



Suppose two hosts are connected by a point-to-point link and they are configured to use Stop-and-Wait protocol for reliable data transfer. Identify in which one of the following scenarios, the utilization of the link is the lowest.

- A. Longer link length and lower transmission rate
- B. Longer link length and higher transmission rate
- C. Shorter link length and lower transmission rate
- D. Shorter link length and higher transmission rate

5

gatecse-2023 computer-networks stop-and-wait one-mark 6

Answer key

2.33.5 Stop and Wait: GATE IT 2005 | Question: 72



A channel has a bit rate of 4 *kbps* and one-way propagation delay of 20 *ms*. The channel uses stop and wait protocol. The transmission time of the acknowledgment frame is negligible. To get a channel efficiency of at least 50%, the minimum frame size should be

- A. 80 bytes
- B. 80 bits
- C. 160 bytes
- D. 160 bits

8

gateit-2005 computer-networks stop-and-wait normal 9

Answer key

2.33.6 Stop and Wait: GATE IT 2006 | Question: 68



On a wireless link, the probability of packet error is 0.2. A stop-and-wait protocol is used to transfer data across the link. The channel condition is assumed to be independent of transmission to transmission. What is the average number of transmission attempts required to transfer 100 packets?

- A. 100
- B. 125
- C. 150
- D. 200

11

gateit-2006 computer-networks sliding-window stop-and-wait normal 12

Answer key 13

2.34

Subnetting (21)

Practice Tests: Test 1 (15Q) Test 2 (13Q)

2.34.1 Subnetting: GATE CSE 2003 | Question: 82, ISRO2009-1



The subnet mask for a particular network is 255.255.31.0. Which of the following pairs of IP addresses could belong to this network?

- A. 172.57.88.62 and 172.56.87.23
- B. 10.35.28.2 and 10.35.29.4
- C. 191.203.31.87 and 191.234.31.88
- D. 128.8.129.43 and 128.8.161.55

gatecse-2003 computer-networks subnetting normal isro2009

Answer key

2.34.2 Subnetting: GATE CSE 2004 | Question: 55



The routing table of a router is shown below: ¹

Destination	Subnet Mask	Interface
128.75.43.0	255.255.255.0	Eth0
128.75.43.0	255.255.255.128	Eth1
192.12.17.5	255.255.255.255	Eth3
Default		Eth2

On which interface will the router forward packets addressed to destinations 128.75.43.16 and 192.12.17.10 respectively? ²

- A. Eth1 and Eth2 B. Eth0 and Eth2 C. Eth0 and Eth3 D. Eth1 and Eth3 ³

gatecse-2004 computer-networks subnetting normal ⁴

Answer key ⁵

2.34.3 Subnetting: GATE CSE 2005 | Question: 27



An organization has a class B network and wishes to form subnets for 64 departments. The subnet mask would be: ⁶

- A. 255.255.0.0 B. 255.255.64.0 C. 255.255.128.0 D. 255.255.252.0 ⁷

gatecse-2005 computer-networks subnetting normal ⁸

Answer key ⁹

2.34.4 Subnetting: GATE CSE 2006 | Question: 45



Two computers C_1 and C_2 are configured as follows. C_1 has IP address 203.197.2.53 and netmask 255.255.128.0. C_2 has IP address 203.197.75.201 and netmask 255.255.192.0. Which one of the following statements is true? ¹⁰

- A. C_1 and C_2 both assume they are on the same network ¹¹
 B. C_2 assumes C_1 is on same network, but C_1 assumes C_2 is on a different network
 C. C_1 assumes C_2 is on same network, but C_2 assumes C_1 is on a different network
 D. C_1 and C_2 both assume they are on different networks.

gatecse-2006 computer-networks subnetting normal ¹²

Answer key ¹³

2.34.5 Subnetting: GATE CSE 2007 | Question: 67, ISRO2016-72



The address of a class B host is to be split into subnets with a 6-bit subnet number. What is the maximum number of subnets and the maximum number of hosts in each subnet? ¹⁴

- A. 62 subnets and 262142 hosts. B. 64 subnets and 262142 hosts. ¹⁵
 C. 62 subnets and 1022 hosts. D. 64 subnets and 1024 hosts.

gatecse-2007 computer-networks subnetting easy isro2016 ¹⁶

Answer key ¹⁷

2.34.6 Subnetting: GATE CSE 2008 | Question: 57



If a class B network on the Internet has a subnet mask of 255.255.248.0, what is the maximum number of hosts per subnet? ¹⁸

- A. 1022 B. 1023 C. 2046 D. 2047 ¹⁹

gatecse-2008 computer-networks subnetting easy ²⁰

Answer key

2.34.7 Subnetting: GATE CSE 2010 | Question: 47



Suppose computers A and B have IP addresses 10.105.1.113 and 10.105.1.91 respectively and they both use same netmask N . Which of the values of N given below should not be used if A and B should belong to the same network?

- A. 255.255.255.0
C. 255.255.255.192
- B. 255.255.255.128
D. 255.255.255.224

gatecse-2010 computer-networks subnetting easy 3

Answer key

2.34.8 Subnetting: GATE CSE 2012 | Question: 34, ISRO-DEC2017-32



An Internet Service Provider (ISP) has the following chunk of CIDR-based IP addresses available with it: 245.248.128.0/20. The ISP wants to give half of this chunk of addresses to Organization A , and a quarter to Organization B , while retaining the remaining with itself. Which of the following is a valid allocation of addresses to A and B ?

- A. 245.248.136.0/21 and 245.248.128.0/22
B. 245.248.128.0/21 and 245.248.128.0/22
C. 245.248.132.0/22 and 245.248.132.0/21
D. 245.248.136.0/24 and 245.248.132.0/21

gatecse-2012 computer-networks subnetting normal isrodec2017 6

Answer key

2.34.9 Subnetting: GATE CSE 2015 | Set 2 | Question: 41



Consider the following routing table at an IP router:

Network No	Net Mask	Next Hop
128.96.170.0	255.255.254.0	Interface 0
128.96.168.0	255.255.254.0	Interface 1
128.96.166.0	255.255.254.0	R2
128.96.164.0	255.255.252.0	R3
0.0.0.0	Default	R4

For each IP address in Group I Identify the correct choice of the next hop from Group II using the entries from the routing table above.

Group I	Group II
i) 128.96.171.92	a) Interface 0
ii) 128.96.167.151	b) Interface 1
iii) 128.96.163.151	c) R2
iv) 128.96.164.121	d) R3
	e) R4

- A. i-a, ii-c, iii-e, iv-d
C. i-b, ii-c, iii-d, iv-e
- B. i-a, ii-d, iii-b, iv-e
D. i-b, ii-c, iii-e, iv-d

gatecse-2015-set2 computer-networks subnetting easy

Answer key

2.34.10 Subnetting: GATE CSE 2015 | Set 3 | Question: 38



In the network 200.10.11.144/27, the fourth octet (in decimal) of the last IP address of the network which can be assigned to a host is _____.

gatecse-2015-set3 computer-networks subnetting normal numerical-answers 2

Answer key

2.34.11 Subnetting: GATE CSE 2019 | Question: 28



Consider three machines M, N, and P with IP addresses 100.10.5.2, 100.10.5.5, and 100.10.5.6 respectively. The subnet mask is set to 255.255.255.252 for all the three machines. Which one of the following is true?

- A. M, N, and P all belong to the same subnet
- B. Only M and N belong to the same subnet
- C. Only N and P belong to the same subnet
- D. M, N, and P belong to three different subnets

gatecse-2019 computer-networks subnetting two-marks 5

Answer key

2.34.12 Subnetting: GATE CSE 2020 | Question: 38



An organization requires a range of IP address to assign one to each of its 1500 computers. The organization has approached an Internet Service Provider (ISP) for this task. The ISP uses CIDR and serves the requests from the available IP address space 202.61.0.0/17. The ISP wants to assign an address space to the organization which will minimize the number of routing entries in the ISP's router using route aggregation. Which of the following address spaces are potential candidates from which the ISP can allot any one of the organization?

- I. 202.61.84.0/21
- II. 202.61.104.0/21
- III. 202.61.64.0/21
- IV. 202.61.144.0/21

- A. I and II only
- B. II and III only
- C. III and IV only
- D. I and IV only

gatecse-2020 computer-networks subnetting two-marks 10

Answer key

2.34.13 Subnetting: GATE CSE 2022 | Question: 45



Consider routing table of an organization's router shown below:

Subnet Number	Subnet Mask	Next Hop
12.20.164.0	255.255.252.0	R1
12.20.170.0	255.255.254.0	R2
12.20.168.0	255.255.254.0	Interface 0
12.20.166.0	255.255.254.0	Interface 1
default		R3

Which of the following prefixes in CIDR notation can be collectively used to correctly aggregate all of the subnets in the routing table?

- A. 12.20.164.0/20
- B. 12.20.164.0/22
- C. 12.20.164.0/21
- D. 12.20.168.0/22

gatecse-2022 computer-networks subnetting multiple-selects two-marks

Answer key

2.34.14 Subnetting: GATE CSE 2023 | Question: 55



The forwarding table of a router is shown below. ¹

Subnet Number	Subnet Mask	Interface ID
200.150.0.0	255.255.0.0	1
200.150.64.0	255.255.224.0	2
200.150.68.0	255.255.255.0	3
200.150.68.64	255.255.255.224	4
Default		0

A packet addressed to a destination address 200.150.68.118 arrives at the router. It will be forwarded to the interface with ID _____. ³

gatecse-2023 computer-networks subnetting numerical-answers two-marks

Answer key ²

2.34.15 Subnetting: GATE CSE 2025 | Set 1 | Question: 30



A packet with the destination IP address 145.36.109.70 arrives at a router whose routing table is shown. Which interface will the packet be forwarded to? ⁴

Subnet Address	Subnet Mask (in CIDR notation)	Interface
145.36.0.0	/16	E1
145.36.128.0	/17	E2
145.36.64.0	/18	E3
145.36.255.0	/24	E4
Default	—	E5

A. E3 B. E1 C. E2 D. E5 ⁶

gatecse2025-set1 computer-networks subnetting two-marks ⁷

Answer key ⁸

2.34.16 Subnetting: GATE IT 2004 | Question: 26



A subnet has been assigned a subnet mask of 255.255.255.192. What is the maximum number of hosts that can belong to this subnet? ⁹

A. 14 B. 30 C. 62 D. 126 ¹⁰

gateit-2004 computer-networks subnetting normal ¹¹

Answer key ¹²

2.34.17 Subnetting: GATE IT 2005 | Question: 76



A company has a class C network address of 204.204.204.0. It wishes to have three subnets, one with 100 hosts and two with 50 hosts each. Which one of the following options represents a feasible set of subnet address/subnet mask pairs?

- A. 204.204.204.128/255.255.255.192
204.204.204.0/255.255.255.128
204.204.204.64/255.255.255.128
- B. 204.204.204.0/255.255.255.192
204.204.204.192/255.255.255.128
204.204.204.64/255.255.255.128
- C. 204.204.204.128/255.255.255.128

204.204.204.192/255.255.255.192
204.204.204.224/255.255.255.192
D. 204.204.204.128/255.255.255.128
204.204.204.64/255.255.255.192
204.204.204.0/255.255.255.192

gateit-2005 computer-networks subnetting normal

Answer key

2.34.18 Subnetting: GATE IT 2006 | Question: 63, ISRO2015-57



A router uses the following routing table: 1

Destination	Mask	Interface
144.16.0.0	255.255.0.0	eth0
144.16.64.0	255.255.224.0	eth1
144.16.68.0	255.255.255.0	eth2
144.16.68.64	255.255.255.224	eth3

Packet bearing a destination address 144.16.68.117 arrives at the router. On which interface will it be forwarded? 3

A. eth0 B. eth1 C. eth2 D. eth3 4

gateit-2006 computer-networks subnetting normal isro2015 5

Answer key 6

2.34.19 Subnetting: GATE IT 2006 | Question: 70



A subnetted Class B network has the following broadcast address: 144.16.95.255 7
Its subnet mask

A. is necessarily 255.255.224.0
B. is necessarily 255.255.240.0
C. is necessarily 255.255.248.0
D. could be any one of 255.255.224.0, 255.255.240.0, 255.255.248.0 8

gateit-2006 computer-networks subnetting normal

Answer key

2.34.20 Subnetting: GATE IT 2008 | Question: 84



Host X has IP address 192.168.1.97 and is connected through two routers R_1 and R_2 to another host Y with IP address 192.168.1.80. Router R_1 has IP addresses 192.168.1.135 and 192.168.1.110. R_2 has IP addresses 192.168.1.67 and 192.168.1.155. The netmask used in the network is 255.255.255.224. 9

Given the information above, how many distinct subnets are guaranteed to already exist in the network? 10

A. 1 B. 2 C. 3 D. 6 11

gateit-2008 computer-networks subnetting normal 12

Answer key

2.34.21 Subnetting: GATE IT 2008 | Question: 85



Host X has IP address 192.168.1.97 and is connected through two routers R_1 and R_2 to another host Y with IP address 192.168.1.80. Router R_1 has IP addresses 192.168.1.135 and 192.168.1.110. R_2 has IP addresses 192.168.1.67 and 192.168.1.155. The netmask used in the network is 255.255.255.224.

Which IP address should X configure its gateway as?

A. 192.168.1.67 B. 192.168.1.110

C. 192.168.1.135

D. 192.168.1.155 1

gateit-2008 computer-networks subnetting normal

Answer key

2.35

TCP (20)

Practice Tests: Test 1 (15Q) Test 2 (15Q) Test 3 (4Q) 2

2.35.1 TCP: GATE CSE 2009 | Question: 47



While opening a *TCP* connection, the initial sequence number is to be derived using a time-of-day (ToD) clock that keeps running even when the host is down. The low order 32 bits of the counter of the ToD clock is to be used for the initial sequence numbers. The clock counter increments once per milliseconds. The maximum packet lifetime is given to be 64s.

Which one of the choices given below is closest to the minimum permissible rate at which sequence numbers used for packets of a connection can increase? 5

A. 0.015/s

B. 0.064/s

C. 0.135/s

D. 0.327/s 6

gatecse-2009 computer-networks tcp difficult ambiguous 7

Answer key

2.35.2 TCP: GATE CSE 2012 | Question: 22



Which of the following transport layer protocols is used to support electronic mail? 8

A. SMTP

B. IP

C. TCP

D. UDP 9

gatecse-2012 computer-networks tcp easy 10

Answer key

2.35.3 TCP: GATE CSE 2015 | Set 1 | Question: 19



Suppose two hosts use a TCP connection to transfer a large file. Which of the following statements is/are FALSE with respect to the TCP connection? 11

- I. If the sequence number of a segment is m , then the sequence number of the subsequent segment is always $m + 1$.
- II. If the estimated round trip time at any given point of time is t sec, the value of the retransmission timeout is always set to greater than or equal to t sec.
- III. The size of the advertised window never changes during the course of the TCP connection.
- IV. The number of unacknowledged bytes at the sender is always less than or equal to the advertised window.

A. III only

B. I and III only

C. I and IV only

D. II and IV only 12

gatecse-2015-set1 computer-networks tcp normal 14

Answer key

2.35.4 TCP: GATE CSE 2015 | Set 2 | Question: 34



Assume that the bandwidth for a TCP connection is 1048560 bits/sec. Let α be the value of RTT in milliseconds (rounded off to the nearest integer) after which the TCP window scale option is needed. Let β be the maximum possible window size with window scale option. Then the values of α and β are

A. 63 milliseconds, 65535×2^{14} B. 63 milliseconds, 65535×2^{16} 17C. 500 milliseconds, 65535×2^{14} D. 500 milliseconds, 65535×2^{16}

gatecse-2015-set2 computer-networks difficult tcp 18

Answer key

2.35.5 TCP: GATE CSE 2015 | Set 3 | Question: 22



Consider the following statements.

- I. TCP connections are full duplex

- II. TCP has no option for selective acknowledgement ¹
III. TCP connections are message streams

- A. Only I is correct
C. Only II and III are correct
- B. Only I and III are correct ²
D. All of I, II and III are correct

gatecse-2015-set3 computer-networks tcp easy

Answer key

2.35.6 TCP: GATE CSE 2016 | Set 2 | Question: 25

Identify the correct sequence in which the following packets are transmitted on the network by a host when a browser requests a webpage from a remote server, assuming that the host has just been restarted. ³

- A. HTTP GET request, DNS query, TCP SYN
C. DNS query, TCP SYN, HTTP GET request.
- B. DNS query, HTTP GET request, TCP SYN ⁴
D. TCP SYN, DNS query, HTTP GET request.

gatecse-2016-set2 computer-networks normal tcp ⁵

Answer key

2.35.7 TCP: GATE CSE 2017 | Set 1 | Question: 14

Consider a TCP client and a TCP server running on two different machines. After completing data transfer, the TCP client calls close to terminate the connection and a FIN segment is sent to the TCP server. Server-side TCP responds by sending an ACK, which is received by the client-side TCP. As per the TCP connection state diagram (RFC 793), in which state does the client-side TCP connection wait for the FIN from the server-side TCP? ⁶

- A. LAST-ACK
B. TIME-WAIT
C. FIN-WAIT-1
D. FIN-WAIT-2 ⁷

gatecse-2017-set1 computer-networks tcp

Answer key

2.35.8 TCP: GATE CSE 2021 | Set 1 | Question: 44

A TCP server application is programmed to listen on port number P on host S . A TCP client is connected to the TCP server over the network. ⁸

Consider that while the TCP connection was active, the server machine S crashed and rebooted. Assume that the client does not use the TCP keepalive timer. Which of the following behaviors is/are possible? ⁹

- A. If the client was waiting to receive a packet, it may wait indefinitely ¹⁰
B. The TCP server application on S can listen on P after reboot
C. If the client sends a packet after the server reboot, it will receive a RST segment
D. If the client sends a packet after the server reboot, it will receive a FIN segment

gatecse-2021-set1 multiple-selects computer-networks tcp two-marks

Answer key

2.35.9 TCP: GATE CSE 2021 | Set 1 | Question: 45

Consider two hosts P and Q connected through a router R . The maximum transfer unit (MTU) value of the link between P and R is 1500 bytes, and between R and Q is 820 bytes. ¹¹

A TCP segment of size 1400 bytes was transferred from P to Q through R , with IP identification value as 0x1234. Assume that the IP header size is 20 bytes. Further, the packet is allowed to be fragmented, i.e., Don't Fragment (DF) flag in the IP header is *not* set by P .

Which of the following statements is/are correct?

- A. Two fragments are created at R and the IP datagram size carrying the second fragment is 620 bytes.
B. If the second fragment is lost, R will resend the fragment with the IP identification value 0x1234.
C. If the second fragment is lost, P is required to resend the whole TCP segment.
D. TCP destination port can be determined by analysing *only* the second fragment.

Answer key

2.35.10 TCP: GATE CSE 2021 | Set 2 | Question: 7



Consider the three-way handshake mechanism followed during TCP connection establishment between hosts P and Q . Let X and Y be two random 32-bit starting sequence numbers chosen by P and Q respectively. Suppose P sends a TCP connection request message to Q with a TCP segment having SYN bit = 1, SEQ number = X , and ACK bit = 0. Suppose Q accepts the connection request. Which one of the following choices represents the information present in the TCP segment header that is sent by Q to P ?

- A. SYN bit = 1, SEQ number = $X + 1$, ACK bit = 0, ACK number = Y , FIN bit = 0
- B. SYN bit = 0, SEQ number = $X + 1$, ACK bit = 0, ACK number = Y , FIN bit = 1
- C. SYN bit = 1, SEQ number = Y , ACK bit = 1, ACK number = $X + 1$, FIN bit = 0
- D. SYN bit = 1, SEQ number = Y , ACK bit = 1, ACK number = X , FIN bit = 0

Answer key

2.35.11 TCP: GATE CSE 2022 | Question: 50



Consider the data transfer using TCP over a 1 Gbps link. Assuming that the maximum segment lifetime (MSL) is set to 60 seconds, the minimum number of bits required for the sequence number field of the TCP header, to prevent the sequence number space from wrapping around during the MSL is _____.

Answer key

2.35.12 TCP: GATE CSE 2023 | Question: 40



Suppose you are asked to design a new reliable byte-stream transport protocol like TCP. This protocol, named myTCP, runs over a 100 Mbps network with Round Trip Time of 150 milliseconds and the maximum segment lifetime of 2 minutes.

Which of the following is/are valid lengths of the Sequence Number field in the myTCP header? 5

- A. 30 bits
- B. 32 bits
- C. 34 bits
- D. 36 bits

Answer key

2.35.13 TCP: GATE CSE 2024 | Set 1 | Question: 19



TCP client P successfully establishes a connection to TCP server Q . Let N_P denote the sequence number in the SYN sent from P to Q . Let N_Q denote the acknowledgement number in the SYN ACK from Q to P . Which of the following statements is/are CORRECT?

- A. The sequence number N_P is chosen randomly by P
- B. The sequence number N_P is always 0 for a new connection
- C. The acknowledgement number N_Q is equal to N_P
- D. The acknowledgement number N_Q is equal to $N_P + 1$

Answer key

2.35.14 TCP: GATE CSE 2025 | Set 1 | Question: 12



Consider the 3-way handshaking protocol for TCP connection establishment. Let the three packets exchanged during the connection establishment be denoted as P_1 , P_2 , and P_3 , in order. Which of the following option(s) is/are TRUE with respect to TCP header flags that are set in the packets?

- A. P3 : SYN = 1, ACK = 1
 C. P2: SYN = 0, ACK = 1
 B. P2 : SYN = 1, ACK = 1
 D. P1: SYN = 1

gatecse2025-set1 computer-networks tcp multiple-selects one-mark 2

Answer key

2.35.15 TCP: GATE CSE 2026 | Set 1 | Question: 8



With respect to a TCP connection between a client and a server, which one of the following statements is true?

- A. The client and server use a two-way handshake mechanism before the start of data transmission
 B. The server cannot initiate closing of the connection before the client initiates closing of the connection
 C. The TCP connection is half-duplex
 D. The client and server can initiate closing of the connection at the same time

gatecse-2026-set1 computer-networks tcp one-mark 5

Answer key

2.35.16 TCP: GATE IT 2004 | Question: 23



Which one of the following statements is FALSE?

- A. TCP guarantees a minimum communication rate
 B. TCP ensures in-order delivery
 C. TCP reacts to congestion by reducing sender window size
 D. TCP employs retransmission to compensate for packet loss

gateit-2004 computer-networks tcp normal 8

Answer key

2.35.17 TCP: GATE IT 2004 | Question: 28



In TCP, a unique sequence number is assigned to each

- A. byte
 B. word
 C. segment
 D. message

gateit-2004 computer-networks tcp easy 11

Answer key

2.35.18 TCP: GATE IT 2007 | Question: 13



Consider the following statements about the timeout value used in TCP.

- The timeout value is set to the RTT (Round Trip Time) measured during TCP connection establishment for the entire duration of the connection.
- Appropriate RTT estimation algorithm is used to set the timeout value of a TCP connection.
- Timeout value is set to twice the propagation delay from the sender to the receiver.

Which of the following choices hold?

- A. (i) is false, but (ii) and (iii) are true
 B. (i) and (iii) are false, but (ii) is true
 C. (i) and (ii) are false, but (iii) is true
 D. (i), (ii) and (iii) are false

gateit-2007 computer-networks tcp normal 16

Answer key

2.35.19 TCP: GATE IT 2007 | Question: 14



Consider a TCP connection in a state where there are no outstanding ACKs. The sender sends two segments back to back. The sequence numbers of the first and second segments are 230 and 290 respectively. The first segment was lost, but the second segment was received correctly by the receiver. Let X be

the amount of data carried in the first segment (in bytes), and Y be the ACK number sent by the receiver. ¹
The values of X and Y (in that order) are

- A. 60 and 290 B. 230 and 291 C. 60 and 231 D. 60 and 230 ²

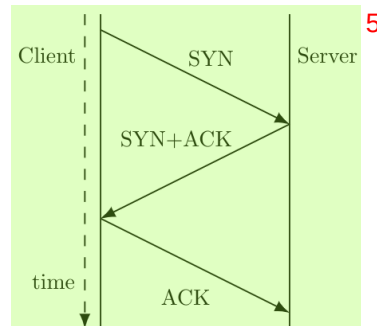
gateit-2007 computer-networks tcp normal ³

Answer key

2.35.20 TCP: GATE IT 2008 | Question: 69



The three way handshake for TCP connection establishment is shown below. ⁴



Which of the following statements are TRUE? ⁶

- $S1$: Loss of $SYN + ACK$ from the server will not establish a connection ⁷
 $S2$: Loss of ACK from the client cannot establish the connection
 $S3$: The server moves $LISTEN \rightarrow SYN_RCVD \rightarrow SYN_SENT \rightarrow ESTABLISHED$ in the state machine on no packet loss
 $S4$: The server moves $LISTEN \rightarrow SYN_RCVD \rightarrow ESTABLISHED$ in the state machine on no packet loss

- A. $S2$ and $S3$ only B. $S1$ and $S4$ only C. $S1$ and $S3$ only D. $S2$ and $S4$ only ⁸

gateit-2008 computer-networks tcp normal

Answer key

2.36

Token Bucket (2)

2.36.1 Token Bucket: GATE CSE 2008 | Question: 58



A computer on a 10Mbps network is regulated by a token bucket. The token bucket is filled at a rate of 2Mbps. It is initially filled to capacity with 16Megabits. What is the maximum duration for which the computer can transmit at the full 10Mbps? ⁹

- A. 1.6 seconds B. 2 seconds C. 5 seconds D. 8 seconds ¹⁰

gatecse-2008 computer-networks token-bucket ¹¹

Answer key

2.36.2 Token Bucket: GATE CSE 2016 | Set 1 | Question: 54



For a host machine that uses the token bucket algorithm for congestion control, the token bucket has a capacity of 1 megabyte and the maximum output rate is 20 megabytes per second. Tokens arrive at a rate to sustain output at a rate of 10 megabytes per second. The token bucket is currently full and the machine needs to send 12 megabytes of data. The minimum time required to transmit the data is _____ seconds.

gatecse-2016-set1 computer-networks token-bucket normal numerical-answers

Answer key

2.37

UDP (4)

2.37.1 UDP: GATE CSE 2005 | Question: 23



Packets of the same session may be routed through different paths in: **1**

- A. TCP, but not UDP
B. TCP and UDP
C. UDP, but not TCP
D. Neither TCP nor UDP

gatecse-2005 computer-networks tcp udp easy **3**

Answer key

2.37.2 UDP: GATE CSE 2013 | Question: 12



The transport layer protocols used for real time multimedia, file transfer, DNS and email, respectively are **4**

- A. TCP, UDP, UDP and TCP
B. UDP, TCP, TCP and UDP
C. UDP, TCP, UDP and TCP
D. TCP, UDP, TCP and UDP

gatecse-2013 computer-networks tcp udp easy **6**

Answer key

2.37.3 UDP: GATE CSE 2017 | Set 2 | Question: 18



Consider socket API on a Linux machine that supports connected UDP sockets. A connected UDP socket is a UDP socket on which connect function has already been called. Which of the following statements is/are CORRECT?

- I. A connected UDP socket can be used to communicate with multiple peers simultaneously. **8**
II. A process can successfully call connect function again for an already connected UDP socket.

- A. I only
B. II only
C. Both I and II
D. Neither I nor II **9**

gatecse-2017-set2 computer-networks udp **10**

Answer key

2.37.4 UDP: GATE IT 2006 | Question: 69



A program on machine X attempts to open a *UDP* connection to port 5376 on a machine Y , and a *TCP* connection to port 8632 on machine Z . However, there are no applications listening at the corresponding ports on Y and Z . An *ICMP* Port Unreachable error will be generated by **11**

- A. Y but not Z
B. Z but not Y
C. Neither Y nor Z
D. Both Y and Z **12**

gateit-2006 computer-networks tcp udp normal **13**

Answer key

2.38

Wrap Around Time (2)

2.38.1 Wrap Around Time: GATE CSE 2014 | Set 3 | Question: 27



Every host in an *IPv4* network has a 1-second resolution real-time clock with battery backup. Each host needs to generate up to 1000 unique identifiers per second. Assume that each host has a globally unique *IPv4* address. Design a 50-bit globally unique ID for this purpose. After what period (in seconds) will the identifiers generated by a host wrap around? **14**

gatecse-2014-set3 computer-networks ip-addressing wrap-around-time numerical-answers normal **15**

Answer key

2.38.2 Wrap Around Time: GATE CSE 2018 | Question: 25



Consider a long-lived *TCP* session with an end-to-end bandwidth of 1 Gbps ($= 10^9$ bits-per-second). The session starts with a sequence number of 1234. The minimum time (in seconds, rounded to the closet integer) before this sequence number can be used again is _____.

gatecse-2018 computer-networks tcp wrap-around-time normal numerical-answers one-mark

Answer key

Answer Keys¹

2.1.1	A	2.1.2	C	2.1.3	C	2.1.4	C	2.1.5	B
2.1.6	6	2.1.7	4	2.1.8	A;C;D	2.1.9	C	2.1.10	B
2.1.11	C	2.1.12	D	2.1.13	C	2.2.1	B	2.3.1	B
2.3.2	D	2.4.1	B	2.4.2	A	2.4.3	A	2.5.1	B
2.5.2	C	2.5.3	C	2.5.4	D	2.5.5	A	2.6.1	D
2.6.2	200	2.6.3	50	2.6.4	C	2.6.5	A	2.6.6	B
2.7.1	A	2.8.1	A	2.8.2	7.07:7.09	2.8.3	B	2.8.4	B
2.9.1	D	2.9.2	C	2.9.3	1100	2.9.4	C	2.9.5	44
2.9.6	4	2.9.7	B	2.9.8	4:4	2.9.9	B	2.10.1	30:30
2.11.1	C	2.11.2	B	2.11.3	A	2.11.4	C	2.11.5	B;C
2.11.6	0.5	2.11.7	B	2.11.8	A	2.12.1	3 : 4	2.12.2	A
2.12.3	C	2.12.4	A	2.12.5	A	2.12.6	B	2.12.7	C
2.12.8	N/A	2.13.1	B	2.13.2	B	2.13.3	D	2.13.4	C
2.13.5	B	2.13.6	500	2.13.7	C	2.14.1	D	2.14.2	B
2.14.3	C	2.14.4	7:7	2.14.5	D	2.15.1	N/A	2.15.2	A
2.16.1	D	2.16.2	C	2.16.3	9	2.16.4	144	2.16.5	C;D
2.16.6	A	2.16.7	D	2.16.8	4094	2.17.1	C	2.17.2	D
2.17.3	D	2.17.4	A	2.17.5	B	2.17.6	C	2.17.7	13
2.17.8	A;C	2.17.9	6	2.17.10	B;C	2.17.11	C;D	2.17.12	D
2.18.1	A	2.19.1	D	2.19.2	A	2.19.3	26	2.19.4	3
2.19.5	D	2.19.6	D	2.19.7	B	2.20.1	D	2.20.2	12
2.20.3	B	2.20.4	C	2.21.1	N/A	2.21.2	D	2.21.3	C
2.21.4	A	2.22.1	D	2.22.2	A	2.22.3	C	2.22.4	B
2.22.5	C	2.22.6	A;B;C	2.23.1	D	2.23.2	C	2.23.3	B
2.23.4	A	2.23.5	C	2.23.6	C	2.23.7	50 : 52	2.23.8	C
2.23.9	C	2.23.10	D	2.23.11	B	2.24.1	D	2.24.2	D
2.24.3	1575	2.24.4	B	2.25.1	B	2.26.1	0.949:0.952	2.27.1	130 : 140
2.28.1	B	2.28.2	D	2.28.3	A	2.28.4	1	2.28.5	C
2.28.6	D	2.28.7	A;C	2.28.8	B	2.28.9	40	2.28.10	B
2.28.11	C	2.28.12	A	2.28.13	C	2.28.14	D	2.29.1	A
2.30.1	B	2.30.2	B	2.30.3	B	2.30.4	C	2.30.5	C
2.30.6	D	2.30.7	C	2.30.8	5	2.30.9	8	2.30.10	4
2.30.11	B	2.30.12	C	2.30.13	B	2.30.14	B	2.30.15	A
2.30.16	A	2.31.1	0.4404	2.32.1	D	2.32.2	C	2.32.3	C
2.32.4	B	2.33.1	320	2.33.2	2500	2.33.3	86.5 : 89.5	2.33.4	B
2.33.5	D	2.33.6	B	2.34.1	D	2.34.2	A	2.34.3	D
2.34.4	C	2.34.5	C	2.34.6	C	2.34.7	D	2.34.8	A
2.34.9	A	2.34.10	158	2.34.11	C	2.34.12	B	2.34.13	B;D

2.34.14	3
2.34.19	D
2.35.3	B
2.35.8	A;B;C
2.35.13	A;D
2.35.18	B
2.37.1	B
2.38.2	34 : 35

1

2.34.15	A
2.34.20	C
2.35.4	C
2.35.9	A;C
2.35.14	B;D
2.35.19	D
2.37.2	C

2

2.34.16	C
2.34.21	B
2.35.5	A
2.35.10	C
2.35.15	D
2.35.20	B
2.37.3	B

3

2.34.17	D
2.35.1	A
2.35.6	C
2.35.11	33
2.35.16	A
2.36.1	B
2.37.4	D

4

2.34.18	C
2.35.2	C
2.35.7	D
2.35.12	B;C;D
2.35.17	A
2.36.2	1.10:1.19
2.38.1	256

5



ER-model. Relational model: Relational algebra, Tuple calculus, SQL. Integrity constraints, Normal forms. File organization, Indexing (e.g., B and B+ trees). Transactions and concurrency control.

Mark Distribution in Previous GATE

Year	2026 - 1	2026 - 2	2025 - 1	2025 - 2	2024 - 1	2024 - 2	2023	2022	2021 - 1	2021 - 2	Minimum
1 Mark Count	2	2	1	1	4	4	1	3	2	1	1
2 Marks Count	2	2	3	4	2	2	2	2	3	3	2
Total Marks	6	6	8	9	8	8	5	7	8	7	5

Welcome to the Databases chapter of your GATE Computer Science preparation. This crucial subject forms the backbone of modern data management and is indispensable for any computer science professional. In GATE, Databases typically carries a significant weightage, ranging from 6 to 10 marks, often featuring a mix of conceptual questions, problem-solving scenarios, and direct application of formulas. You can expect questions on topics like relational algebra/calculus, SQL queries, functional dependencies, normal forms, B-trees, and concurrency control protocols. A strong grasp of these fundamentals is vital not just for the exam but also for practical applications in software development and data engineering.

Topic-wise Key Concepts

Armstrong Axioms

Armstrong Axioms are a set of inference rules used to derive all functional dependencies (FDs) implied by a given set of FDs. They are sound (do not generate incorrect FDs) and complete (can generate all correct FDs).

Formulas/Theorems:

- Reflexivity:** If $B \subseteq A$, then $A \rightarrow B$. (Trivial FD)
- Augmentation:** If $A \rightarrow B$, then $AC \rightarrow BC$ (where C is any set of attributes).
- Transitivity:** If $A \rightarrow B$ and $B \rightarrow C$, then $A \rightarrow C$.

Derived Rules:

- Union:** If $A \rightarrow B$ and $A \rightarrow C$, then $A \rightarrow BC$.
- Decomposition:** If $A \rightarrow BC$, then $A \rightarrow B$ and $A \rightarrow C$.
- Pseudotransitivity:** If $A \rightarrow B$ and $BC \rightarrow D$, then $AC \rightarrow D$.

Key Properties/Identities:

- Used to find the closure of an attribute set X^+ .
- Fundamental for determining candidate keys and normal forms.

Common Pitfalls:

- Confusing derived rules with basic axioms.
- Incorrectly applying augmentation or transitivity.

Problem-Solving Techniques:

- To find X^+ : Start with X . Repeatedly add attributes to X^+ if they are determined by any subset of X^+ using the given FDs.
- To check if $A \rightarrow B$ is implied by a set of FDs F : Check if $B \subseteq A^+$.

B Tree

A B-tree is a self-balancing tree data structure that maintains sorted data and allows searches, sequential access, insertions, and deletions in logarithmic time. It is optimized for systems that read and write large blocks of data, such as disk storage.

Formulas/Theorems:1

- **Order m :** Each node (except root) has at least $\lceil m/2 \rceil$ children and at most m children.2
- **Number of keys:** A node with k children has $k - 1$ keys.
- **Minimum keys (non-root):** $\lceil m/2 \rceil - 1$.
- **Maximum keys:** $m - 1$.
- **Maximum height h :** For N keys and order m , $h \leq \log_{\lceil m/2 \rceil} \left(\frac{N+1}{2} \right)$.
- **Minimum height h :** For N keys and order m , $h \geq \log_m (N + 1)$.

Key Properties/Identities:3

- All leaf nodes are at the same level.4
- Keys within a node are sorted.
- Efficient for disk-based indexing due to high fan-out.

Common Pitfalls:5

- Miscalculating minimum/maximum number of keys or children per node.6
- Incorrectly performing split/merge operations during insertion/deletion.

Problem-Solving Techniques:7

- Trace insertion/deletion operations step-by-step, paying attention to splits (when a node overflows) and merges (when a node underflows)8
- Calculate disk I/Os by counting the number of nodes visited (including root and leaf) for search, insertion, or deletion.

Candidate Key9

A Candidate Key is a minimal Super Key. It is a set of attributes that uniquely identifies each tuple in a relation, and no proper subset of these attributes can uniquely identify tuples.10

Formulas/Theorems:11

- A set of attributes K is a Candidate Key if:12
 1. $K \rightarrow R$ (where R is all attributes in the relation) - Uniqueness property.
 2. For no proper subset $K' \subset K$, $K' \rightarrow R$ - Minimality property.

Key Properties/Identities:13

- Every relation must have at least one candidate key.14
- One candidate key is chosen as the Primary Key.

Common Pitfalls:15

- Confusing with Super Key (Super Key doesn't require minimality).16
- Forgetting to check the minimality condition.

Problem-Solving Techniques:17

- Find the closure of various attribute sets. A set K is a super key if K^+ contains all attributes of the relation.18
- From the super keys, identify those that are minimal (i.e., no subset is also a super key) to find candidate keys.

Conflict Serializable19

A schedule is Conflict Serializable if it is conflict equivalent to some serial schedule. Two operations conflict if they are on the same data item, belong to different transactions, and at least one of them is a write operation (W-W, W-R, R-W conflicts).20

Formulas/Theorems: 1

• Precedence Graph (Serialization Graph): 2

- Nodes represent transactions.
- An edge $T_i \rightarrow T_j$ exists if T_i performs an operation that conflicts with an operation of T_j , and T_i 's operation occurs before T_j 's.
- **Theorem:** A schedule is conflict serializable if and only if its precedence graph contains no cycles.

Key Properties/Identities: 4

- Guarantees that the concurrent execution produces the same result as some sequential execution.
- A weaker condition than view serializability but easier to test.

Common Pitfalls: 6

- Incorrectly identifying conflicting operations (e.g., R-R operations do not conflict).
- Missing cycles in the precedence graph, especially in complex schedules.

Problem-Solving Techniques: 8

- Construct the precedence graph by identifying all conflicting pairs of operations and drawing edges accordingly.
- Use a cycle detection algorithm (e.g., DFS) on the graph to check for serializability.

Database Design 10

Database design is the process of creating a detailed data model for a database. It involves translating conceptual requirements into a logical schema, and then into a physical schema, ensuring data integrity, efficiency, and scalability.

Formulas/Theorems: 12

- No direct formulas, but relies on principles of normalization and ER-to-Relational mapping rules.

Key Properties/Identities: 14

- Phases: Conceptual (ER Model), Logical (Relational Schema), Physical (Storage details).
- Goals: Minimize redundancy, maximize integrity, optimize performance.

Common Pitfalls: 16

- Poor choice of primary/foreign keys.
- Not adequately normalizing the schema, leading to update anomalies.
- Over-normalization, leading to complex queries and performance issues.

Problem-Solving Techniques: 18

- Start with an ER diagram to capture requirements.
- Apply ER-to-Relational mapping rules systematically.
- Normalize relations to an appropriate normal form (e.g., BCNF or 3NF) using functional dependencies.

Database Normalization 20

Database Normalization is a systematic process of restructuring a relational database schema to minimize data redundancy and improve data integrity. It involves decomposing relations into smaller, well-structured relations based on functional dependencies.

Formulas/Theorems: 22

- **1NF (First Normal Form):** All attributes must be atomic (no multi-valued or composite attributes).
- **2NF (Second Normal Form):** In 1NF and no non-prime attribute is partially dependent on a candidate key.
- **3NF (Third Normal Form):** In 2NF and no non-prime attribute is transitively dependent on a candidate key.

- **BCNF (Boyce-Codd Normal Form):** For every non-trivial functional dependency $A \rightarrow B$, A must be a super key. **1**
- **4NF (Fourth Normal Form):** In BCNF and no non-trivial multi-valued dependencies exist.
- **5NF (Fifth Normal Form):** In 4NF and no non-trivial join dependencies exist.

Key Properties/Identities: **2**

- Hierarchy: $1NF \subset 2NF \subset 3NF \subset BCNF \subset 4NF \subset 5NF$. **3**
- Desirable properties of decomposition: Lossless Join and Dependency Preservation. BCNF decomposition is always lossless but not always dependency preserving. 3NF decomposition is always lossless and dependency preserving.

Common Pitfalls: **4**

- Incorrectly identifying prime and non-prime attributes. **5**
- Mistaking partial dependency for transitive dependency.
- Failing to check for both lossless join and dependency preservation after decomposition.

Problem-Solving Techniques: **6**

- Identify all candidate keys and functional dependencies. **7**
- Systematically check for violations of each normal form.
- If a violation exists, decompose the relation into smaller relations.

Database Schema **8**

A Database Schema is the logical structure or design of the entire database. It defines the tables, attributes, data types, relationships, and constraints that govern the data within the database. It is defined using Data Definition Language (DDL). **9**

Formulas/Theorems: **10**

- No specific formulas, but it's the blueprint for the database. **11**

Key Properties/Identities: **12**

- **Internal Schema:** Physical storage structure. **13**
- **Conceptual Schema:** Logical structure of the entire database.
- **External Schema (View):** User-specific views of the database.
- Schema defines the intension, while data (instance) defines the extension.

Common Pitfalls: **14**

- Confusing schema (structure) with database instance (actual data). **15**
- Overlooking important constraints during schema design.

Problem-Solving Techniques: **16**

- Understand the mapping from ER diagrams to relational schemas. **17**
- Be familiar with DDL commands like CREATE TABLE, ALTER TABLE, DROP TABLE.

Decomposition **18**

Decomposition is the process of breaking down a relation (table) into two or more smaller relations. It is primarily used in normalization to eliminate redundancy and anomalies, ensuring that the resulting relations are in a higher normal form. **19**

Formulas/Theorems: **20**

- **Lossless Join Decomposition:** A decomposition of R into R_1 and R_2 is lossless if $R = R_1 \bowtie R_2$. **21**
 - Condition: $(R_1 \cap R_2) \rightarrow R_1$ or $(R_1 \cap R_2) \rightarrow R_2$ must hold based on the FDs of R .
- **Dependency Preservation:** A decomposition preserves dependencies if the union of FDs on the decomposed

relations is equivalent to the original set of FDs. Formally, if F is the original set of FDs and F_i is the set of FDs on R_i , then $(F_1 \cup F_2 \cup \dots \cup F_k)^+ = F^+$. 1

Key Properties/Identities: 2

- Desirable properties for decomposition are lossless join and dependency preservation. 3
- BCNF decomposition is always lossless but not always dependency preserving.
- 3NF decomposition is always lossless and dependency preserving.

Common Pitfalls: 4

- Incorrectly applying the lossless join test. 5
- Failing to check for dependency preservation, especially for BCNF.

Problem-Solving Techniques: 6

- For lossless join: Check if the common attributes form a super key for at least one of the decomposed relations. 7
- For dependency preservation: For each original FD $A \rightarrow B$, check if it can be derived from the FDs of the decomposed relations.

ER Diagram 8

An Entity-Relationship (ER) Diagram is a high-level conceptual data model that represents the entities, attributes, and relationships within a system. It is used during the conceptual design phase of database development. 9

Formulas/Theorems: 10

- No direct formulas, but specific rules for mapping to relational schema. 11

Key Properties/Identities: 12

- **Entities:** Represented by rectangles (e.g., Student, Course). 13
- **Attributes:** Represented by ovals (e.g., Name, ID). Key attributes are underlined.
- **Relationships:** Represented by diamonds (e.g., Enrolls, Teaches).
- **Cardinality:** (1:1, 1:N, M:N) specifies the number of instances of one entity that can be associated with another.
- **Participation:** (Total/Partial) specifies whether an entity instance must participate in a relationship.
- **Weak Entity:** An entity that cannot be uniquely identified by its own attributes and depends on a strong (owner) entity. Represented by double rectangles.

Common Pitfalls: 14

- Incorrectly identifying cardinalities or participation constraints. 15
- Confusing strong and weak entities.
- Errors in mapping complex ER constructs (e.g., ternary relationships, generalization) to relational schema.

Problem-Solving Techniques: 16

- Carefully read the problem description to identify entities, attributes, and relationships. 17
- Pay close attention to cardinality and participation rules.
- Practice mapping ER diagrams to relational schemas using standard rules.

Functional Dependency 18

A Functional Dependency (FD) $A \rightarrow B$ means that the value of attribute set A uniquely determines the value of attribute set B . If two tuples have the same values for attributes in A , they must also have the same values for attributes in B . 19

Formulas/Theorems: 20

- **Closure of an attribute set X^+ :** The set of all attributes that are functionally determined by X . 21

- Algorithm: Start with $X^+ = X$. Repeatedly add attributes Y to X^+ if there is an FD $W \rightarrow Y$ such that $W \subseteq X^+$. 1

- **Minimal Cover (or Minimal Basis):** A set of FDs F_m equivalent to F such that: 2
 1. Every FD in F_m has a single attribute on the right-hand side.
 2. No FD in F_m can be removed without changing F_m^+ .
 3. No attribute can be removed from the left-hand side of any FD in F_m without changing F_m^+ .

Key Properties/Identities: 3

- Fundamental for database normalization and identifying keys. 4
- Armstrong's Axioms are used to infer FDs.

Common Pitfalls: 5

- Incorrectly calculating attribute closure. 6
- Errors in finding a minimal cover.

Problem-Solving Techniques: 7

- Master the algorithm for finding X^+ . 8
- For minimal cover, systematically apply the three conditions: right-hand side single attribute, remove redundant FDs, remove redundant attributes from LHS.

Indexing 9

Indexing is a technique used to optimize the performance of database queries by allowing the database server to quickly locate and retrieve specific rows without scanning the entire table. It creates a data structure (like B-tree or hash table) that stores a small, ordered subset of the data. 10

Formulas/Theorems: 11

- **Disk I/O for B-tree index:** For a search, typically height of tree + 1 (for data block) disk accesses. 12
- **Disk I/O for hash index:** Typically 1-2 disk accesses for exact match, if no collision.

Key Properties/Identities: 13

- **Primary Index:** Index on the primary key, usually clustered. 14
- **Secondary Index:** Index on non-key attributes, usually unclustered.
- **Clustered Index:** Data rows are stored physically in the order of the index key. Only one per table.
- **Unclustered Index:** Index stores pointers to the data rows, which are not physically ordered by the index key.

Common Pitfalls: 15

- Misunderstanding the difference between clustered and unclustered indexes. 16
- Incorrectly calculating disk I/Os for different index types and query patterns.

Problem-Solving Techniques: 17

- Analyze the query type (equality search, range search) and index type to determine disk I/O. 18
- Remember that clustered indexes can retrieve multiple data records with fewer I/Os for range queries.

Joins 19

Joins combine rows from two or more tables based on a related column between them. They are fundamental operations in relational algebra and SQL for retrieving data from multiple relations. 20

Formulas/Theorems: 21

- **Cartesian Product (Cross Join):** $R \times S$. Combines every row of R with every row of S . If R has n tuples and S 22

has m tuples, $R \times S$ has $n \times m$ tuples. ¹

- **Theta Join:** $R \bowtie_{\theta} S = \sigma_{\theta}(R \times S)$. Combines tuples from R and S where the condition θ is true. ²
- **Equijoin:** A theta join where θ is an equality condition (e.g., $R.A = S.B$).
- **Natural Join:** $R \bowtie S$. An equijoin on all common attributes, with duplicate common columns removed from the result.
- **Outer Joins (Left, Right, Full):** Preserve tuples that do not have a match in the other relation, filling unmatched attributes with NULLs.

Key Properties/Identities: ³

- Result schema of a join is the union of attributes from participating relations (with common attributes appearing once in natural join). ⁴
- Join operations are associative and commutative (for inner joins).

Common Pitfalls: ⁵

- Confusing different types of joins, especially natural join vs. equijoin. ⁶
- Misunderstanding how NULL values are handled in outer joins.
- Incorrectly calculating the number of tuples in the result.

Problem-Solving Techniques: ⁷

- For natural join, identify common attributes and their values. ⁸
- For outer joins, remember to include unmatched tuples from the specified side(s).
- Trace the operation step-by-step for small example relations.

Multivalued Dependency 4NF ⁹

A Multivalued Dependency (MVD) $A \twoheadrightarrow B$ exists in a relation R if, for a given value of A , there is a set of values for B , ¹⁰ and this set is independent of the values of other attributes in R . 4NF (Fourth Normal Form) requires that for every non-trivial MVD $A \twoheadrightarrow B$ in a relation, A must be a super key.

Formulas/Theorems: ¹¹

- **Trivial MVD:** $A \twoheadrightarrow B$ is trivial if $B \subseteq A$ or $A \cup B = R$. ¹²
- **MVD Inference Rules (some are similar to FDs):**
 - **Reflexivity:** If $B \subseteq A$, then $A \twoheadrightarrow B$.
 - **Augmentation:** If $A \twoheadrightarrow B$, then $AC \twoheadrightarrow BC$.
 - **Transitivity:** If $A \twoheadrightarrow B$ and $B \twoheadrightarrow C$, then $A \twoheadrightarrow (C - B)$.
 - **Complementation:** If $A \twoheadrightarrow B$, then $A \twoheadrightarrow (R - A - B)$.
 - **Union:** If $A \twoheadrightarrow B$ and $A \twoheadrightarrow C$, then $A \twoheadrightarrow BC$.
 - **Decomposition:** If $A \twoheadrightarrow B$ and $A \twoheadrightarrow C$, then $A \twoheadrightarrow (B \cap C)$, $A \twoheadrightarrow (B - C)$, and $A \twoheadrightarrow (C - B)$.
 - **Replication (FD to MVD):** If $A \rightarrow B$, then $A \twoheadrightarrow B$.

Key Properties/Identities: ¹³

- MVDs address redundancy arising from multi-valued facts that are independent of each other. ¹⁴
- 4NF eliminates these MVDs by decomposition.

Common Pitfalls: ¹⁵

- Confusing MVDs with FDs. MVDs imply FDs, but not vice versa. ¹⁶
- Incorrectly identifying trivial MVDs.

Problem-Solving Techniques: ¹⁷

- Identify MVDs by looking for independent multi-valued facts. ¹⁸
- Decompose relations based on non-trivial MVDs to achieve 4NF. For $R(A, B, C)$ with $A \twoheadrightarrow B$, decompose into $R_1(A, B)$ and $R_2(A, C)$.

Natural Join¹

The Natural Join operation ($\bowtie$) combines two relations based on equality of their common attributes. It implicitly creates an equijoin condition for all attributes that share the same name in both relations and projects out the duplicate common attributes.²

Formulas/Theorems:³

- Given relations R and S , let $C = \text{attributes}(R) \cap \text{attributes}(S)$ be the set of common attributes.⁴

$$R \bowtie S = \pi_{\text{attributes}(R) \cup \text{attributes}(S)} (\sigma_{R.c_1=S.c_1 \wedge \dots \wedge R.c_k=S.c_k} (R \times S))$$

where $c_i \in C$.

Key Properties/Identities:⁵

- The resulting schema contains all attributes from both relations, with common attributes appearing only once.⁶
- It is a special case of equijoin followed by projection.

Common Pitfalls:⁷

- Misunderstanding the implicit join condition (all common attributes).⁸
- Incorrectly determining the schema or number of tuples in the result.

Problem-Solving Techniques:⁹

- Identify all common attributes between the two relations.¹⁰
- For each pair of tuples (one from each relation), check if their values for all common attributes are equal. If so, combine them into a single result tuple, keeping common attributes only once.

Normal Forms¹¹

Normal Forms (1NF, 2NF, 3NF, BCNF, 4NF, 5NF) are a series of guidelines for designing relational database schemas to reduce data redundancy and improve data integrity. Each normal form builds upon the previous one, imposing stricter rules.¹²

Formulas/Theorems:¹³

- 1NF:** All attributes are atomic.¹⁴
- 2NF:** In 1NF and no non-prime attribute is partially dependent on any candidate key.
- 3NF:** In 2NF and no non-prime attribute is transitively dependent on any candidate key.
- BCNF:** For every non-trivial FD $A \rightarrow B$, A is a super key.
- 4NF:** In BCNF and no non-trivial MVDs exist.
- 5NF:** In 4NF and no non-trivial join dependencies exist.

Key Properties/Identities:¹⁵

- Higher normal forms generally reduce more redundancy but may require more joins for queries.¹⁶
- BCNF is generally preferred, but 3NF is often a practical compromise as it guarantees lossless join and dependency preservation.

Common Pitfalls:¹⁷

- Incorrectly identifying candidate keys, which is crucial for all normal forms.¹⁸
- Confusing partial, transitive, and multi-valued dependencies.

Problem-Solving Techniques:¹⁹

- First, find all candidate keys and the closure of all FDs.²⁰
- Then, systematically check the conditions for 1NF, 2NF, 3NF, and BCNF in order.

- If a relation is not in a desired normal form, decompose it into smaller relations that satisfy the conditions. 1

Query 2

A query is a request for data or information from a database. Queries are typically written using a query language like SQL, Relational Algebra, or Relational Calculus. They specify what data to retrieve, how to filter it, and how to present it. 3

Formulas/Theorems: 4

- No specific formulas, but relies on the syntax and semantics of query languages. 5

Key Properties/Identities: 6

- **Selectivity:** The fraction of tuples that satisfy a selection condition. 7
- **Projection:** Selecting specific columns.
- **Join:** Combining data from multiple tables.

Common Pitfalls: 8

- Syntax errors in SQL. 9
- Logical errors leading to incorrect results (e.g., wrong join conditions, incorrect aggregation).
- Inefficient queries that perform poorly.

Problem-Solving Techniques: 10

- Break down complex queries into smaller, manageable parts. 11
- Understand the order of operations in SQL (FROM, WHERE, GROUP BY, HAVING, SELECT, ORDER BY).
- Practice translating natural language requirements into formal query language expressions.

Referential Integrity 12

Referential Integrity is a database concept that ensures that relationships between tables remain consistent. It is enforced using foreign key constraints, which dictate that a foreign key in one table must either match a primary key in another table or be NULL. 13

Formulas/Theorems: 14

- No specific formulas, but it's a constraint rule. 15

Key Properties/Identities: 16

- Prevents "dangling references" where a foreign key refers to a non-existent primary key. 17
- Actions on deletion/update of primary key: CASCADE, SET NULL, SET DEFAULT, RESTRICT (default).

Common Pitfalls: 18

- Violating referential integrity constraints during data modification. 19
- Misunderstanding the behavior of different ON DELETE/ON UPDATE actions.

Problem-Solving Techniques: 20

- Identify primary and foreign keys correctly during schema design. 21
- Choose appropriate ON DELETE/ON UPDATE actions based on business rules.
- Trace data modification operations to check for constraint violations.

Relational Algebra 22

Relational Algebra is a procedural query language that takes relations as input and produces relations as output. It consists of a set of fundamental operations (selection, projection, union, set difference, Cartesian product, rename) and 23

derived operations (join, intersection, division).1

Formulas/Theorems:2

- **Select:** $\sigma_P(R)$ (selects tuples satisfying predicate P).3
- **Project:** $\pi_A(R)$ (selects attributes A).
- **Union:** $R \cup S$ (combines tuples from R and S ; relations must be union-compatible).
- **Set Difference:** $R \setminus S$ (tuples in R but not in S ; union-compatible).
- **Cartesian Product:** $R \times S$ (combines every tuple of R with every tuple of S).
- **Rename:** $\rho_{S(A_1, \dots, A_n)}(R)$ (renames relation R to S and its attributes).
- **Intersection:** $R \cap S = R \setminus (R \setminus S)$ (derived).
- **Join:** $R \bowtie_P S = \sigma_P(R \times S)$ (derived).

Key Properties/Identities:4

- Closure property: The result of any operation is also a relation.5
- Forms the theoretical basis for SQL.

Common Pitfalls:6

- Incorrectly applying operators, especially for complex queries.7
- Forgetting union compatibility requirements for set operations.
- Operator precedence issues.

Problem-Solving Techniques:8

- Break down complex queries into a sequence of simpler relational algebra operations.9
- Practice converting SQL queries to relational algebra and vice versa.

Relational Calculus10

Relational Calculus is a non-procedural (declarative) query language that describes what data to retrieve without specifying how to retrieve it. It comes in two forms: Tuple Relational Calculus (TRC) and Domain Relational Calculus (DRC).11

Formulas/Theorems:12

- **Tuple Relational Calculus (TRC):** $\{t \mid P(t)\}$ where t is a tuple variable and $P(t)$ is a formula (predicate) involving t .13
 - Example: $\{t \mid t \in \text{Student} \wedge t.\text{Age} > 20\}$
- **Domain Relational Calculus (DRC):** $\{ \langle x_1, x_2, \dots, x_n \rangle \mid P(x_1, x_2, \dots, x_n) \}$ where x_i are domain variables and P is a formula.
 - Example: $\{ \langle N, A \rangle \mid \exists I (\langle I, N, A \rangle \in \text{Student} \wedge A > 20) \}$

Key Properties/Identities:14

- Expressive power equivalent to Relational Algebra (Codd's Theorem).15
- Uses quantifiers ($\exists$ - existential, $\forall$ - universal).

Common Pitfalls:16

- Incorrectly using quantifiers, especially universal quantification.17
- Formulating predicates that are not "safe" (may yield infinite results).

Problem-Solving Techniques:18

- Translate natural language queries into logical predicates using tuple or domain variables.19
- Pay close attention to the scope of quantifiers.
- Ensure queries are safe by bounding all variables.

Relational Model¹

The Relational Model is a data model based on the concept of relations (tables). Data is organized into two-dimensional tables, where each table represents an entity or a relationship, and rows (tuples) represent records, while columns (attributes) represent fields.²

Formulas/Theorems:³

- No specific formulas, but defines the structure.⁴

Key Properties/Identities:⁵

- **Relation (Table):** A set of tuples.⁶
- **Tuple (Row):** A record in a relation.
- **Attribute (Column):** A named property of a relation.
- **Domain:** The set of permissible values for an attribute.
- **Schema:** The logical design of the database.
- **Instance:** The actual data stored in the database at a particular time.
- **Integrity Constraints:** Entity Integrity (Primary Key not NULL), Referential Integrity (Foreign Key rules).

Common Pitfalls:⁷

- Confusing the formal definitions of relation, tuple, and attribute.⁸
- Not understanding the role of domains and integrity constraints.

Problem-Solving Techniques:⁹

- Understand the basic terminology and how data is represented.¹⁰
- Be able to identify components of a relational schema.

SQL¹¹

SQL (Structured Query Language) is the standard language for managing and manipulating relational databases. It is used for defining database schemas (DDL), querying data (DML), controlling access (DCL), and managing transactions (TCL).¹²

Formulas/Theorems:¹³

- No specific formulas, but adheres to a strict syntax.¹⁴

Key Properties/Identities:¹⁵

- **DDL (Data Definition Language):** CREATE, ALTER, DROP (for schema objects).¹⁶
- **DML (Data Manipulation Language):** SELECT, INSERT, UPDATE, DELETE (for data).
- **DCL (Data Control Language):** GRANT, REVOKE (for permissions).
- **TCL (Transaction Control Language):** COMMIT, ROLLBACK, SAVEPOINT.

Common Pitfalls:¹⁷

- Complex queries involving subqueries, aggregate functions, GROUP BY, and HAVING clauses.¹⁸
- Understanding the order of execution of SQL clauses.
- Differences in SQL dialects (though GATE usually sticks to standard SQL).

Problem-Solving Techniques:¹⁹

- Practice writing queries for various scenarios, including joins, subqueries, and aggregation.²⁰
- Understand the logical processing order of a SELECT statement: FROM -> WHERE -> GROUP BY -> HAVING -> SELECT -> ORDER BY.

Safe Query²¹

A query in Relational Calculus is considered "safe" if it produces a finite number of tuples as its result for any valid database instance. Unsafe queries can potentially produce an infinite result set, which is undesirable in practice. 1

Formulas/Theorems: 2

- No direct formulas, but a conceptual property. 3

Key Properties/Identities: 4

- Every query expressible in Relational Algebra is safe. 5
- Safety typically requires that all variables are "bounded" – their values must come from the active domain of the database (values currently present in the tables).

Common Pitfalls: 6

- Queries with unbounded variables, especially those using universal quantifiers without proper restrictions. 7
- Example of unsafe query: $\{t \mid \neg(t \in R)\}$ (all tuples not in R, potentially infinite).

Problem-Solving Techniques: 8

- Ensure that all variables in a relational calculus query are restricted to range over finite sets (e.g., relations in the database or active domain). 9
- For universal quantifiers, ensure they are used in a limited context (e.g., "for all X in Y...").

Super Key 10

A Super Key is any set of attributes in a relation that, taken together, uniquely identifies each tuple in that relation. It guarantees uniqueness but does not necessarily imply minimality. 11

Formulas/Theorems: 12

- A set of attributes K is a Super Key if $K \rightarrow R$ (where R is all attributes in the relation). 13

Key Properties/Identities: 14

- Every candidate key is a super key. 15
- A super key can contain redundant attributes (i.e., attributes not necessary for uniqueness).

Common Pitfalls: 16

- Confusing with Candidate Key (Candidate Key is a minimal super key). 17
- Forgetting that adding more attributes to a super key still results in a super key.

Problem-Solving Techniques: 18

- To find super keys, start with candidate keys and add any combination of remaining attributes. 19
- Alternatively, find the closure of attribute sets. Any set X for which X^+ contains all attributes of the relation is a super key.

Timestamp Ordering 20

Timestamp Ordering is a concurrency control protocol that ensures serializability by assigning a unique timestamp to each transaction. Transactions are executed in the order of their timestamps, and conflicts are resolved by rolling back transactions that violate this order. 21

Formulas/Theorems: 22

- Each transaction T_i is assigned a unique timestamp $TS(T_i)$. 23
- Each data item X has a $R_TS(X)$ (read timestamp) and a $W_TS(X)$ (write timestamp).
- **Read Rule:** If $TS(T_i) < W_TS(X)$, T_i must abort and restart with a new timestamp. Otherwise, read X and set

$$R_TS(X) = \max(R_TS(X), TS(T_i)).$$

- **Write Rule:** If $TS(T_i) < R_TS(X)$ or $TS(T_i) < W_TS(X)$, T_i must abort and restart. Otherwise, write X and set $W_TS(X) = TS(T_i)$.

Key Properties/Identities:2

- Guarantees conflict serializability.
- Can lead to cascading rollbacks and starvation.
- Basic timestamp ordering is prone to aborts.

Common Pitfalls:4

- Incorrectly applying the read/write rules, especially the abort conditions.
- Tracing complex schedules with multiple transactions.

Problem-Solving Techniques:6

- Assign timestamps to transactions.
- Maintain $R_TS(X)$ and $W_TS(X)$ for each data item.
- Step through the schedule, applying the read and write rules and determining if any transaction needs to abort.

Transaction and Concurrency8

A Transaction is a logical unit of work that accesses and possibly modifies the contents of a database. Concurrency Control is the management of simultaneous operations on a database to ensure data consistency and integrity, especially when multiple transactions execute concurrently.

Formulas/Theorems:10

- No direct formulas, but concepts like serializability and isolation levels are key.

Key Properties/Identities:12

- **ACID Properties of Transactions:**
 - **Atomicity:** All or nothing.
 - **Consistency:** Transaction takes database from one consistent state to another.
 - **Isolation:** Concurrent transactions appear to execute serially.
 - **Durability:** Changes are permanent once committed.
- **Concurrency Anomalies:** Dirty Read, Lost Update, Unrepeatable Read, Phantom Read.
- **Isolation Levels:** Read Uncommitted, Read Committed, Repeatable Read, Serializable.

Common Pitfalls:15

- Identifying which anomaly occurs in a given schedule.
- Understanding how different concurrency control mechanisms (locking, timestamping) prevent anomalies.

Problem-Solving Techniques:17

- Analyze schedules to identify potential anomalies.
- Understand how each ACID property is maintained or violated.
- Relate isolation levels to the anomalies they prevent.

Tuple Relational Calculus19

Tuple Relational Calculus (TRC) is a non-procedural query language where variables range over tuples. It allows users to specify the properties of the tuples they want in the result without detailing the exact steps to retrieve them.

Formulas/Theorems:21

- General form: $\{t \mid P(t)\}$ where t is a tuple variable and $P(t)$ is a formula (predicate) that t must satisfy.

- Predicates can involve: 1
 - Membership: $t \in R$
 - Attribute access: $t.A$
 - Comparison operators: $=, \neq, <, >, \leq, \geq$
 - Logical connectives: $\wedge, \vee, \neg$
 - Quantifiers: $\exists$ (exists), $\forall$ (for all)

Key Properties/Identities: 2

- Expressive power equivalent to Relational Algebra. 3
- More declarative than Relational Algebra.

Common Pitfalls: 4

- Incorrectly formulating complex predicates, especially with multiple tuple variables and quantifiers. 5
- Ensuring the query is "safe" (produces a finite result).

Problem-Solving Techniques: 6

- Break down the query into smaller logical conditions. 7
- Use a separate tuple variable for each relation involved in the query.
- Carefully use quantifiers, ensuring variables are appropriately bound.

Two Phase Locking Protocol 8

The Two-Phase Locking (2PL) protocol is a concurrency control mechanism that ensures serializability by requiring transactions to acquire all necessary locks before releasing any. It divides a transaction's execution into two phases: a growing phase and a shrinking phase. 9

Formulas/Theorems: 10

- No direct formulas, but a protocol with rules for lock acquisition and release. 11

Key Properties/Identities: 12

- **Growing Phase:** Transaction can acquire new locks but cannot release any. 13
- **Shrinking Phase:** Transaction can release existing locks but cannot acquire new ones.
- Guarantees conflict serializability.
- Can lead to deadlocks.
- **Strict 2PL:** All exclusive (write) locks are held until commit/rollback, preventing cascading rollbacks.

Common Pitfalls: 14

- Incorrectly identifying the growing and shrinking phases. 15
- Detecting deadlocks in schedules using 2PL (e.g., using a wait-for graph).
- Understanding the difference between basic 2PL and strict 2PL.

Problem-Solving Techniques: 16

- Trace transaction execution, noting when locks are acquired and released. 17
- Construct a wait-for graph to detect deadlocks: an edge $T_i \rightarrow T_j$ exists if T_i is waiting for a lock held by T_j . A cycle indicates a deadlock.

Quick Formula Reference 18

- **Armstrong Axioms:** 19
 - Reflexivity: If $B \subseteq A$, then $A \rightarrow B$ 20
 - Augmentation: If $A \rightarrow B$, then $AC \rightarrow BC$
 - Transitivity: If $A \rightarrow B$ and $B \rightarrow C$, then $A \rightarrow C$
- **B-Tree (Order m):** 21

- Min keys (non-root): $\lceil m/2 \rceil - 1$
- Max keys: $m - 1$
- Min children (non-root): $\lceil m/2 \rceil$
- Max children: m
- **Candidate Key:** $K \rightarrow R$ and K is minimal.
- **Conflict Serializable:** Precedence graph has no cycles.
- **Functional Dependency Closure:** X^+ (algorithm based on Armstrong's axioms).
- **Decomposition Lossless Join:** For R_1, R_2 , $(R_1 \cap R_2) \rightarrow R_1$ or $(R_1 \cap R_2) \rightarrow R_2$.
- **Relational Algebra Operators:**
 - Select: $\sigma_P(R)$
 - Project: $\pi_A(R)$
 - Union: $R \cup S$
 - Set Difference: $R \setminus S$
 - Cartesian Product: $R \times S$
 - Natural Join: $R \bowtie S$
- **Tuple Relational Calculus:** $\{t \mid P(t)\}$
- **Timestamp Ordering Rules (for transaction T_i on data item X):**
 - Read: If $TS(T_i) < W_TS(X)$, abort T_i . Else, $R_TS(X) = \max(R_TS(X), TS(T_i))$.
 - Write: If $TS(T_i) < R_TS(X)$ or $TS(T_i) < W_TS(X)$, abort T_i . Else, $W_TS(X) = TS(T_i)$.

1

Important Tips for GATE 2

- **Master Functional Dependencies and Normal Forms:** These are high-yield topics. Practice finding candidate keys, attribute closures, and checking normal forms (especially 3NF and BCNF). Understand the implications of lossless join and dependency preservation.
- **Hands-on with Relational Algebra/Calculus and SQL:** Be proficient in writing queries in all three languages. Practice converting queries between them. Pay attention to operator precedence in RA and quantifier usage in RC.
- **B-Tree Operations and I/O Calculation:** Understand the structure, insertion, deletion, and search operations. Crucially, be able to calculate the number of disk I/Os for various scenarios, as this is a common numerical question.
- **Concurrency Control Protocols:** Understand the ACID properties, common anomalies (dirty read, lost update, etc.), and how 2PL and Timestamp Ordering prevent them. Practice drawing precedence graphs for conflict serializability and wait-for graphs for deadlock detection.
- **ER Diagram to Relational Schema Mapping:** Know the standard rules for mapping entities, attributes (simple, composite, multi-valued), and relationships (1:1, 1:N, M:N, weak entities) to relational tables.
- **Read Questions Carefully:** Database questions often have subtle details. For example, in normalization, the given FDs are critical. In concurrency, the exact sequence of operations matters.
- **Time Management:** Some problems, like tracing B-tree operations or complex concurrency schedules, can be time-consuming. Practice efficiently solving these to save time during the exam.
- **Conceptual Clarity:** While formulas are important, a deep conceptual understanding of why certain rules exist (e.g., why BCNF is stricter than 3NF, why 2PL guarantees serializability) will help you tackle tricky theoretical questions.

3

3.1

Armstrong Axioms (1)

3.1.1 Armstrong Axioms: GATE CSE 2026 | Set 2 | Question: 32



In the context of schema normalization in relational DBMS, consider a set F of functional dependencies. The set of all functional dependencies implied by F is called the closure of F . To compute the closure of F , Armstrong's Axioms can be applied. Consider X, Y , and Z as sets of attributes over a relational schema. The three rules of Armstrong's Axioms are described as follows.

Reflexivity: If $Y \subseteq X$, then $X \rightarrow Y$

Augmentation: If $X \rightarrow Y$, then $XZ \rightarrow YZ$ for any Z

Transitivity: If $X \rightarrow Y$ and $Y \rightarrow Z$, then $X \rightarrow Z$

6

The additional rule of Union is defined as follows.

Union: If $X \rightarrow Y$ and $X \rightarrow Z$, then $X \rightarrow YZ$

8

It can be proved that the additional rule of Union is also implied by the three rules of Armstrong's Axioms. Listed below are four combinations of these three rules. Which one of these combinations is both necessary and sufficient for the proof?

9

A. Reflexivity, Augmentation, and Transitivity
C. Transitivity

B. Reflexivity and Augmentation
D. Augmentation and Transitivity

gatecse-2026-set2 databases database-normalization armstrong-axioms two-marks

Answer key

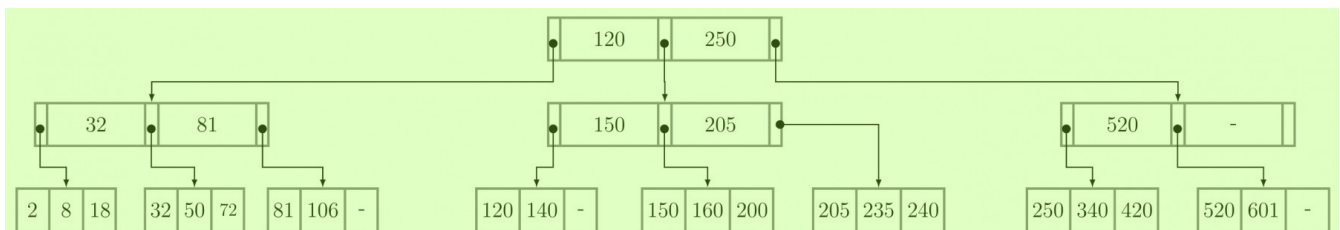
3.2 B Tree (32)

Practice Tests: Test 1 (15Q) Test 2 (15Q) Test 3 (9Q)

3.2.1 B Tree: GATE CSE 1989 | Question: 12a



The below figure shows a B^+ tree where only key values are indicated in the records. Each block can hold upto three records. A record with a key value 34 is inserted into the B^+ tree. Obtain the modified B^+ tree after insertion.



descriptive gate1989 databases b-tree

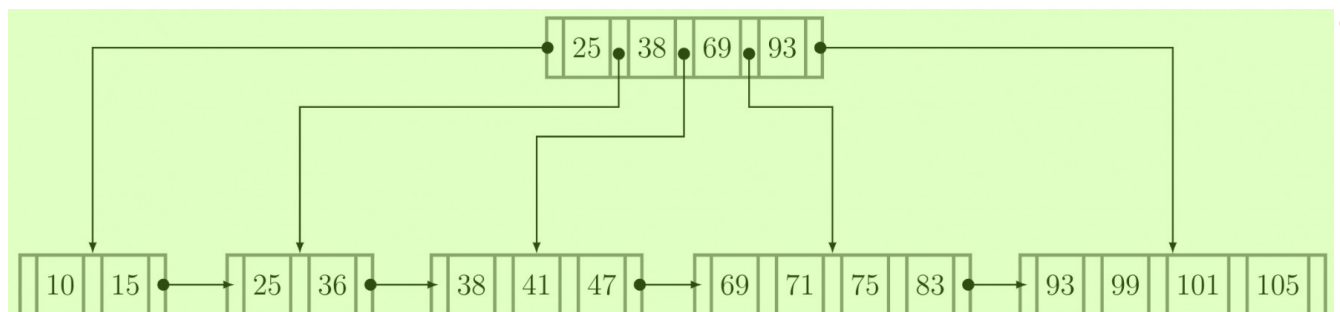
Answer key

3.2.2 B Tree: GATE CSE 1994 | Question: 14a



Consider B^+ - tree of order d shown in figure. (A B^+ - tree of order d contains between d and $2d$ keys in each node)

Draw the resulting B^+ - tree after 100 is inserted in the figure below.



gate1994 databases b-tree normal descriptive

Answer key

3.2.3 B Tree: GATE CSE 1994 | Question: 14b



For a B^+ - tree of order d with n leaf nodes, the number of nodes accessed during a search is $O(_)$.

gate1994 databases b-tree normal descriptive

Answer key

3.2.4 B Tree: GATE CSE 1997 | Question: 19



A B^+ - tree of order d is a tree in which each internal node has between d and $2d$ key values. An internal node with M key values has $M + 1$ children. The root (if it is an internal node) has between 1 and $2d$ key values. The distance of a node from the root is the length of the path from the root to the node. All leaves are at the same distance from the root. The height of the tree is the distance of a leaf from the root.

- A. What is the total number of key values in the internal nodes of a B^+ -tree with l leaves ($l \geq 2$)? **1**
- B. What is the maximum number of internal nodes in a B^+ - tree of order 4 with 52 leaves?
- C. What is the minimum number of leaves in a B^+ -tree of order d and height h ($h \geq 1$)?

gate1997 databases b-tree normal descriptive **2**

Answer key

3.2.5 B Tree: GATE CSE 1999 | Question: 1.25

Which of the following is correct? **3**

- A. B-trees are for storing data on disk and B^+ trees are for main memory. **4**
- B. Range queries are faster on B^+ trees.
- C. B-trees are for primary indexes and B^+ trees are for secondary indexes.
- D. The height of a B^+ tree is independent of the number of records.

gate1999 databases b-tree normal

Answer key

3.2.6 B Tree: GATE CSE 1999 | Question: 21

Consider a B-tree with degree m , that is, the number of children, c , of any internal node (except the root) is such that $m \leq c \leq 2m - 1$. Derive the maximum and minimum number of records in the leaf nodes for such a B-tree with height h , $h \geq 1$. (Assume that the root of a tree is at height 0).

gate1999 databases b-tree normal descriptive **7**

Answer key

3.2.7 B Tree: GATE CSE 2000 | Question: 1.22, UGCNET-June2012-II: 11

B^+ -trees are preferred to binary trees in databases because **9**

- A. Disk capacities are greater than memory capacities **10**
- B. Disk access is much slower than memory access
- C. Disk data transfer rates are much less than memory data transfer rates
- D. Disks are more reliable than memory

gatecse-2000 databases b-tree normal ugcnetcse-june2012-paper2 **11**

Answer key

3.2.8 B Tree: GATE CSE 2000 | Question: 21

(a) Suppose you are given an empty B^+ tree where each node (leaf and internal) can store up to 5 key values. Suppose values $1, 2, \dots, 10$ are inserted, in order, into the tree. Show the tree pictorially **12**

- i. after 6 insertions, and **14**
- ii. after all 10 insertions

Do NOT show intermediate stages. **15**

(b) Suppose instead of splitting a node when it is full, we try to move a value to the left sibling. If there is no left sibling, or the left sibling is full, we split the node. Show the tree after values $1, 2, \dots, 9$ have been inserted. Assume, as in (a) that each node can hold up to 5 keys. **16**

(c) In general, suppose a B^+ tree node can hold a maximum of m keys, and you insert a long sequence of keys in increasing order. Then what approximately is the average number of keys in each leaf level node. **17**

- i. in the normal case, and **18**
- ii. with the insertion as in (b).

Answer key

3.2.9 B Tree: GATE CSE 2001 | Question: 22



We wish to construct a B^+ tree with fan-out (the number of pointers per node) equal to 3 for the following set of key values:

80, 50, 10, 70, 30, 100, 90 ²

Assume that the tree is initially empty and the values are added in the order given. ³

- Show the tree after insertion of 10, after insertion of 30, and after insertion of 90. Intermediate trees need not be shown. ⁴
- The key values 30 and 10 are now deleted from the tree in that order show the tree after each deletion.

Answer key

3.2.10 B Tree: GATE CSE 2002 | Question: 17



- The following table refers to search items for a key in B -trees and B^+ trees. ⁵

B-tree		B ⁺ -tree	
Successful search	Unsuccessful search	Successful search	Unsuccessful search
X_1	X_2	X_3	X_4

⁶

A successful search means that the key exists in the database and unsuccessful means that it is not present in the database. Each of the entries X_1, X_2, X_3 and X_4 can have a value of either Constant or Variable. Constant means that the search time is the same, independent of the specific key value, where variable means that it is dependent on the specific key value chosen for the search. ⁷

Give the correct values for the entries X_1, X_2, X_3 and X_4 (for example $X_1 = \text{Constant}$, $X_2 = \text{Constant}$, $X_3 = \text{Constant}$, $X_4 = \text{Constant}$)

- Relation $R(A, B)$ has the following view defined on it: ⁸

```
CREATE VIEW V AS
(SELECT R1.A, R2.B
FROM R AS R1, R AS R2
WHERE R1.B=R2.A)
```

⁹

- The current contents of relation R are shown below. What are the contents of the view V ? ¹⁰

A	B
1	2
2	3
2	4
4	5
6	7
6	8
9	10

¹¹

- The tuples (2, 11) and (11, 6) are now inserted into R . What are the *additional* tuples that are inserted in V ? ¹²

Answer key

3.2.11 B Tree: GATE CSE 2002 | Question: 2.23, UGCNET-June2012-II: 26



A B^+ - tree index is to be built on the *Name* attribute of the relation *STUDENT*. Assume that all the student names are of length 8 bytes, disk blocks are of size 512 bytes, and index pointers are of size 4 bytes. Given the scenario, what would be the best choice of the degree (i.e. number of pointers per node) of the B^+ - tree?

- A. 16 B. 42 C. 43 D. 44 ²

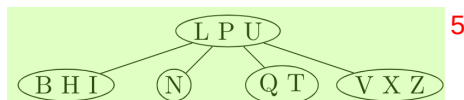
gatescse-2002 databases b-tree normal ugcnetcse-june2012-paper2 ³

Answer key

3.2.12 B Tree: GATE CSE 2003 | Question: 65

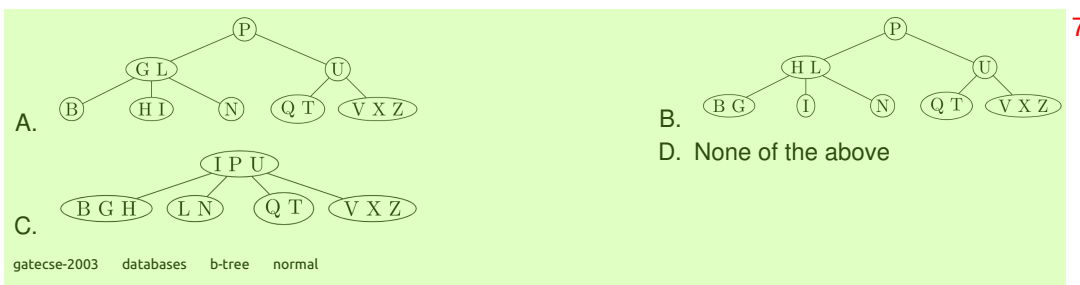


Consider the following 2 — 3 — 4 tree (i.e., B-tree with a minimum degree of two) in which each data item is a letter. The usual alphabetical ordering of letters is used in constructing the tree.



⁵

What is the result of inserting *G* in the above tree? ⁶



⁷

gatescse-2003 databases b-tree normal

Answer key

3.2.13 B Tree: GATE CSE 2004 | Question: 52



The order of an internal node in a B^+ tree index is the maximum number of children it can have. Suppose that a child pointer takes 6 bytes, the search field value takes 14 bytes, and the block size is 512 bytes. What is the order of the internal node?

- A. 24 B. 25 C. 26 D. 27 ⁹

gatescse-2004 databases b-tree normal ¹⁰

Answer key ¹¹

3.2.14 B Tree: GATE CSE 2005 | Question: 28



Which of the following is a key factor for preferring B^+ -trees to binary search trees for indexing database relations? ¹²

- A. Database relations have a large number of records ¹⁴
 B. Database relations are sorted on the primary key
 C. B^+ -trees require less memory than binary search trees
 D. Data transfer from disks is in blocks

gatescse-2005 databases b-tree normal ¹⁵

Answer key ¹⁶

3.2.15 B Tree: GATE CSE 2007 | Question: 63, ISRO2016-59



The order of a leaf node in a B^+ - tree is the maximum number of (value, data record pointer) pairs it can hold. Given that the block size is 1K bytes, data record pointer is 7 bytes long, the value field is 9 bytes long and a block pointer is 6 bytes long, what is the order of the leaf node?

A. 63 B. 64 C. 67 D. 68 ¹

gatecse-2007 databases b-tree normal isro2016 ²

Answer key 

3.2.16 B Tree: GATE CSE 2008 | Question: 41



A B-tree of order 4 is built from scratch by 10 successive insertions. What is the maximum number of nodes splitting operations that may take place? ³

A. 3 B. 4 C. 5 D. 6 ⁴

gatecse-2008 databases b-tree normal ⁵

Answer key  ⁶

3.2.17 B Tree: GATE CSE 2009 | Question: 44



The following key values are inserted into a B^+ - tree in which order of the internal nodes is 3, and that of the leaf nodes is 2, in the sequence given below. The order of internal nodes is the maximum number of tree pointers in each node, and the order of leaf nodes is the maximum number of data items that can be stored in it. The B^+ - tree is initially empty ⁷

10, 3, 6, 8, 4, 2, 1 ⁸

The maximum number of times leaf nodes would get split up as a result of these insertions is ⁹

A. 2 B. 3 C. 4 D. 5 ¹⁰

gatecse-2009 databases b-tree normal ¹¹

Answer key  ¹²

3.2.18 B Tree: GATE CSE 2010 | Question: 18



Consider a B^+ -tree in which the maximum number of keys in a node is 5. What is the minimum number of keys in any non-root node? ¹³

A. 1 B. 2 C. 3 D. 4 ¹⁴

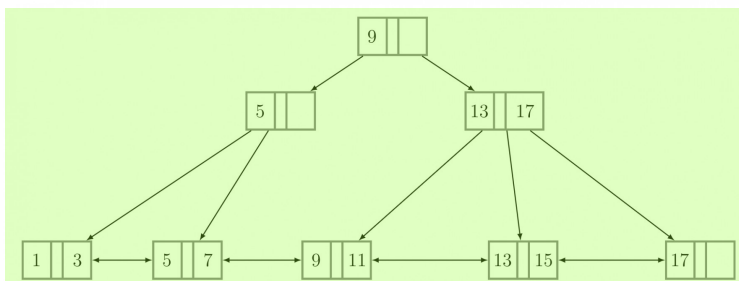
gatecse-2010 databases b-tree easy ¹⁵

Answer key  ¹⁶

3.2.19 B Tree: GATE CSE 2015 | Set 2 | Question: 6



With reference to the B^+ tree index of order 1 shown below, the minimum number of nodes (including the Root node) that must be fetched in order to satisfy the following query. "Get all records with a search key greater than or equal to 7 and less than 15" is _____. ¹⁷



¹⁸

gatecse-2015-set2 databases b-tree normal numerical-answers

Answer key 

3.2.20 B Tree: GATE CSE 2015 | Set 3 | Question: 46



Consider a B^+ tree in which the search key is 12 bytes long, block size is 1024 bytes, record pointer is 10 bytes long and the block pointer is 8 bytes long. The maximum number of keys that can be accommodated

in each non-leaf node of the tree is _____. 1

gatecse-2015-set3 databases b-tree normal numerical-answers 2

Answer key

3.2.21 B Tree: GATE CSE 2016 | Set 2 | Question: 21



B+ Trees are considered BALANCED because. 3

- A. The lengths of the paths from the root to all leaf nodes are all equal. 4
- B. The lengths of the paths from the root to all leaf nodes differ from each other by at most 1.
- C. The number of children of any two non-leaf sibling nodes differ by at most 1.
- D. The number of records in any two leaf nodes differ by at most 1.

gatecse-2016-set2 databases b-tree normal 5

Answer key

3.2.22 B Tree: GATE CSE 2017 | Set 2 | Question: 49



In a B^+ Tree, if the search-key value is 8 bytes long, the block size is 512 bytes and the pointer size is 2 B, then the maximum order of the B^+ Tree is _____

gatecse-2017-set2 databases b-tree numerical-answers normal 7

Answer key

3.2.23 B Tree: GATE CSE 2019 | Question: 14



Which one of the following statements is NOT correct about the B^+ tree data structure used for creating an index of a relational database table? 8

- A. B^+ Tree is a height-balanced tree 9
- B. Non-leaf nodes have pointers to data records
- C. Key values in each node are kept in sorted order
- D. Each leaf node has a pointer to the next leaf node

gatecse-2019 databases b-tree one-mark

Answer key

3.2.24 B Tree: GATE CSE 2024 | Set 1 | Question: 11



In a B^+ tree, the requirement of at least half-full (50%) node occupancy is relaxed for which one of the following cases? 10

- A. Only the root node
- B. All leaf nodes 11
- C. All internal nodes
- D. Only the leftmost leaf node

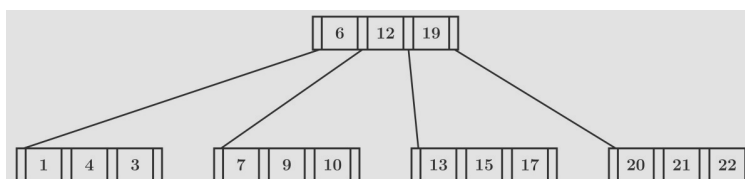
gatecse-2024-set1 databases b-tree one-mark 12

Answer key

3.2.25 B Tree: GATE CSE 2025 | Set 1 | Question: 11



Consider the following B^+ tree with 5 nodes, in which a node can store at most 3 key values. The value 23 is now inserted in the B^+ tree. Which of the following options(s) is/are CORRECT?



- A. None of the nodes will split.
- B. At least one node will split and redistribute.

- C. The total number of nodes will remain same. ¹
 D. The height of the tree will increase.

gatecse2025-set1 databases b-tree multiple-selects one-mark

Answer key 

3.2.26 B Tree: GATE CSE 2025 | Set 2 | Question: 47

In a B^+ -tree where each node can hold at most four key values, a root to leaf path consists of the following nodes: ²

$$A = (49, 77, 83, -), B = (7, 19, 33, 44), C = (20^*, 22^*, 25^*, 26^*)$$
 ³

The *-marked keys signify that these are data entries in a leaf. ⁴

Assume that a pointer between keys k_1 and k_2 points to a subtree containing keys in $[k_1, k_2)$, and that when a leaf is created, the smallest key in it is copied up into its parent. ⁵

A record with key value 23 is inserted into the B^+ -tree. ⁶

The smallest key value in the parent of the leaf that contains 25^* is _____. (Answer in integer) ⁷

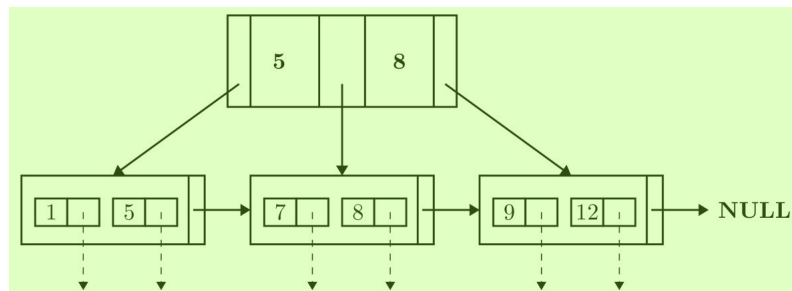
gatecse2025-set2 databases b-tree numerical-answers two-marks ⁸

Answer key 

3.2.27 B Tree: GATE DA 2026 | Question: 31

Consider a B+ Tree where the maximum number of key values in each leaf node is 2 and the maximum number of pointers in each non-leaf node is 3. Let the content of the B+ Tree be as shown in the figure. ⁹

Which of the following options denotes the key value(s) stored in the root node after inserting a key value 3 in the given B+ Tree? ¹⁰



- A. 5 B. 8 C. 3 and 5 D. 3, 5 and 8 ¹¹

gateda-2026 databases b-tree two-marks ¹²

Answer key ¹³

3.2.28 B Tree: GATE IT 2004 | Question: 79

Consider a table T in a relational database with a key field K . A B-tree of order p is used as an access structure on K , where p denotes the maximum number of tree pointers in a B-tree index node. Assume that K is 10 bytes long; disk block size is 512 bytes; each data pointer P_D is 8 bytes long and each block pointer P_B is 5 bytes long. In order for each B-tree node to fit in a single disk block, the maximum value of p is ¹⁴

- A. 20 B. 22 C. 23 D. 32 ¹⁵

gateit-2004 databases b-tree normal

Answer key 

3.2.29 B Tree: GATE IT 2005 | Question: 23, ISRO2017-67



A B-Tree used as an index for a large database table has four levels including the root node. If a new key is inserted in this index, then the maximum number of nodes that could be newly created in the process are

- A. 5 B. 4 C. 3 D. 2

gateit-2005 databases b-tree normal isro2017 3

Answer key

3.2.30 B Tree: GATE IT 2006 | Question: 61



In a database file structure, the search key field is 9 bytes long, the block size is 512 bytes, a record pointer is 7 bytes and a block pointer is 6 bytes. The largest possible order of a non-leaf node in a B^+ tree implementing this file structure is

- A. 23 B. 24 C. 34 D. 44

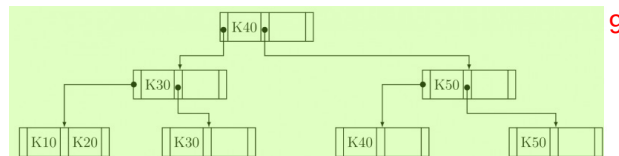
gateit-2006 databases b-tree normal 7

Answer key

3.2.31 B Tree: GATE IT 2007 | Question: 84



Consider the B^+ tree in the adjoining figure, where each node has at most two keys and three links. 8



Keys K_{15} and then K_{25} are inserted into this tree in that order. Exactly how many of the following nodes (disregarding the links) will be present in the tree after the two insertions? 10

- | | |
|-----|-----|
| K30 | K50 |
|-----|-----|

K25	K30
-----	-----

K20	K25
-----	-----

K15	K20
-----	-----

 11

- A. 1 B. 2 C. 3 D. 4

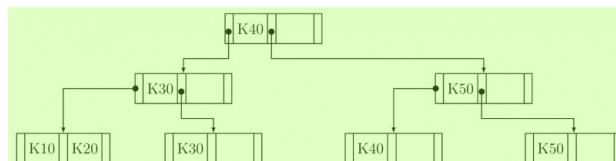
gateit-2007 databases b-tree normal 13

Answer key

3.2.32 B Tree: GATE IT 2007 | Question: 85



Consider the B^+ tree in the adjoining figure, where each node has at most two keys and three links. 14



Keys K_{15} and then K_{25} are inserted into this tree in that order. Now the key K_{50} is deleted from the B^+ tree resulting after the two insertions made earlier. Consider the following statements about the B^+ tree resulting after this deletion. 16

- i. The height of the tree remains the same.
ii. The node

K20		
-----	--	--

 (disregarding the links) is present in the tree.
iii. The root node remains unchanged (disregarding the links).

Which one of the following options is true? 18

- A. Statements (i) and (ii) are true B. Statements (ii) and (iii) are true
C. Statements (iii) and (i) are true D. All the statements are false

gateit-2007 databases b-tree normal 20

Answer key

3.3

Candidate Key (7)

Practice Test: Test 1 (6Q)

3.3.1 Candidate Key: GATE CSE 1994 | Question: 3.7

An instance of a relational scheme $R(A, B, C)$ has distinct values for attribute A . Can you conclude that A is a candidate key for R ?

gate1994 databases easy database-normalization candidate-key descriptive 2

Answer key

3.3.2 Candidate Key: GATE CSE 2005 | Question: 78

Consider a relation scheme $R = (A, B, C, D, E, H)$ on which the following functional dependencies hold: $\{A \rightarrow B, BC \rightarrow D, E \rightarrow C, D \rightarrow A\}$. What are the candidate keys for R ?

- A. AE, BE B. AE, BE, DE C. AEH, BEH, BCH D. AEH, BEH, DEH 4

gatecse-2005 databases candidate-key easy 5

Answer key

3.3.3 Candidate Key: GATE CSE 2011 | Question: 126

Consider a relational table with a single record for each registered student with the following attributes: 7

1. **Registration_Num**: Unique registration number for each registered student
 2. **UID**: Unique identity number, unique at the national level for each citizen
 3. **BankAccount_Num**: Unique account number at the bank. A student can have multiple accounts or joint accounts. This attribute stores the primary account number.
 4. **Name**: Name of the student
 5. **Hostel_Room**: Room number of the hostel
- 8

Which of the following options is **INCORRECT**? 9

- A. **BankAccount_Num** is a candidate key 10
B. **Registration_Num** can be a primary key
C. **UID** is a candidate key if all students are from the same country
D. If S is a super key such that $S \cap \text{UID}$ is NULL then $S \cup \text{UID}$ is also a superkey

gatecse-2011 databases normal candidate-key

Answer key

3.3.4 Candidate Key: GATE CSE 2014 | Set 2 | Question: 21

The maximum number of superkeys for the relation schema $R(E, F, G, H)$ with E as the key is _____. 11

gatecse-2014-set2 databases numerical-answers easy candidate-key 12

Answer key

3.3.5 Candidate Key: GATE CSE 2014 | Set 2 | Question: 22

Given an instance of the STUDENTS relation as shown as below

StudentID	StudentName	StudentEmail	StudentAge	CPI
2345	Shankar	shankar@math	X	9.4
1287	Swati	swati@ee	19	9.5
7853	Shankar	shankar@cse	19	9.4
9876	Swati	swati@mech	18	9.3
8765	Ganesh	ganesh@civil	19	8.7

1

For (StudentName, StudentAge) to be a key for this instance, the value X should NOT be equal to _____. 2

gatecse-2014-set2 databases numerical-answers easy candidate-key 3

Answer key

3.3.6 Candidate Key: GATE CSE 2014 | Set 3 | Question: 22

A *prime attribute* of a relation scheme R is an attribute that appears 4

- A. in all candidate keys of R
- B. in some candidate key of R 5
- C. in a foreign key of R
- D. only in the primary key of R

gatecse-2014-set3 databases easy candidate-key 6

Answer key

3.3.7 Candidate Key: GATE DA 2026 | Question: 7

Let $R(A, B, C, D, E)$ be a relational schema with functional dependency set $F = \{A \rightarrow BC, CD \rightarrow E, E \rightarrow A\}$.

Which of the following statements is correct? 8

- A. AD , ED and CD are the only candidate keys of R . 9
- B. AD and ED are the only candidate keys of R .
- C. A , E and CD are the only candidate keys of R .
- D. A and CD are the only candidate keys of R .

gateda-2026 databases functional-dependency candidate-key one-mark 10

Answer key

3.4 Conflict Serializable (12)

Practice Tests: Test 1 (15Q) Test 2 (4Q) 11

3.4.1 Conflict Serializable: GATE CSE 2007 | Question: 64

Consider the following schedules involving two transactions. Which one of the following statements is TRUE? 12

- $S_1 : r_1(X); r_1(Y); r_2(X); r_2(Y); w_2(Y); w_1(X)$ 13
- $S_2 : r_1(X); r_2(X); r_2(Y); w_2(Y); r_1(Y); w_1(X)$

- A. Both S_1 and S_2 are conflict serializable. 14
- B. S_1 is conflict serializable and S_2 is not conflict serializable.
- C. S_1 is not conflict serializable and S_2 is conflict serializable.
- D. Both S_1 and S_2 are not conflict serializable.

gatecse-2007 databases transaction-and-concurrency normal conflict-serializable

Answer key

3.4.2 Conflict Serializable: GATE CSE 2009 | Question: 43



Consider two transactions T_1 and T_2 , and four schedules S_1, S_2, S_3, S_4 , of T_1 and T_2 as given below.¹

$T_1 : R_1[x]W_1[x]W_1[y]$
 $T_2 : R_2[x]R_2[y]W_2[y]$
 $S_1 : R_1[x]R_2[x]R_2[y]W_1[x]W_1[y]W_2[y]$
 $S_2 : R_1[x]R_2[x]R_2[y]W_1[x]W_2[y]W_1[y]$
 $S_3 : R_1[x]W_1[x]R_2[x]W_1[y]R_2[y]W_2[y]$
 $S_4 : R_2[x]R_2[y]R_1[x]W_1[x]W_1[y]W_2[y]$

Which of the above schedules are conflict-serializable?³

- A. S_1 and S_2 B. S_2 and S_3 C. S_3 only D. S_4 only⁴

gatecse-2009 databases transaction-and-concurrency normal conflict-serializable⁵

Answer key

3.4.3 Conflict Serializable: GATE CSE 2014 | Set 1 | Question: 29



Consider the following four schedules due to three transactions (indicated by the subscript) using *read* and *write* on a data item x , denoted by $r(x)$ and $w(x)$ respectively. Which one of them is conflict serializable?⁷

- A. $r_1(x); r_2(x); w_1(x); r_3(x); w_2(x);$
 B. $r_2(x); r_1(x); w_2(x); r_3(x); w_1(x);$
 C. $r_3(x); r_2(x); r_1(x); w_2(x); w_1(x);$
 D. $r_2(x); w_2(x); r_3(x); r_1(x); w_1(x);$ ⁸

gatecse-2014-set1 databases transaction-and-concurrency conflict-serializable normal

Answer key

3.4.4 Conflict Serializable: GATE CSE 2014 | Set 2 | Question: 29



Consider the following schedule **S** of transactions T_1, T_2, T_3, T_4 :⁹

T1	T2	T3	T4
Writes(X) Commit	Reads(X) Writes(Y) Reads(Z) Commit	Writes(X) Commit	Reads(X) Reads(Y) Commit

Which one of the following statements is CORRECT?¹¹

- A. **S** is conflict-serializable but not recoverable
 B. **S** is not conflict-serializable but is recoverable
 C. **S** is both conflict-serializable and recoverable
 D. **S** is neither conflict-serializable nor is it recoverable¹²

Answer key

3.4.5 Conflict Serializable: GATE CSE 2014 | Set 3 | Question: 29

Consider the transactions T_1, T_2 , and T_3 and the schedules S_1 and S_2 given below. ¹

- $T_1 : r_1(X); r_1(Z); w_1(X); w_1(Z)$ ²
- $T_2 : r_2(Y); r_2(Z); w_2(Z)$
- $T_3 : r_3(Y); r_3(X); w_3(Y)$
- $S_1 : r_1(X); r_3(Y); r_3(X); r_2(Y); r_2(Z); w_3(Y); w_2(Z); r_1(Z); w_1(X); w_1(Z)$
- $S_2 : r_1(X); r_3(Y); r_2(Y); r_3(X); r_1(Z); r_2(Z); w_3(Y); w_1(X); w_2(Z); w_1(Z)$

Which one of the following statements about the schedules is **TRUE**? ³

- A. Only S_1 is conflict-serializable.
- B. Only S_2 is conflict-serializable.
- C. Both S_1 and S_2 are conflict-serializable.
- D. Neither S_1 nor S_2 is conflict-serializable. ⁴

Answer key

3.4.6 Conflict Serializable: GATE CSE 2017 | Set 2 | Question: 44

Two transactions T_1 and T_2 are given as ⁵

$T_1 : r_1(X)w_1(X)r_1(Y)w_1(Y)$ ⁶

$T_2 : r_2(Y)w_2(Y)r_2(Z)w_2(Z)$

where $r_i(V)$ denotes a *read* operation by transaction T_i on a variable V and $w_i(V)$ denotes a *write* operation by transaction T_i on a variable V . The total number of conflict serializable schedules that can be formed by T_1 and T_2 is ⁷

Answer key

3.4.7 Conflict Serializable: GATE CSE 2020 | Question: 37

Consider a schedule of transactions T_1 and T_2 : ⁹

T_1	RA			RC		WD		WB	Commit	
T_2		RB	WB		RD		WC			Commit

¹⁰

Here, RX stands for “Read(X)” and WX stands for “Write(X)”. Which one of the following schedules is conflict equivalent to the above schedule? ¹¹

A. ¹²

T_1				RA	RC	WD	WB		Commit	
T_2	RB	WB	RD					WC		Commit

¹³

B. ¹⁴

T_1	RA	RC	WD	WB					Commit	
T_2					RB	WB	RD	WC		Commit

¹⁵

C. ¹⁶

T_1	RA	RC	WD				WB		Commit	
T_2				RB	WB	RD		WC		Commit

¹⁷

D. ¹⁸

T_1					RA	RC	WD	WB	Commit	
T_2	RB	WB	RD	WC						Commit

Answer key

3.4.8 Conflict Serializable: GATE CSE 2021 | Set 1 | Question: 32

Let $r_i(z)$ and $w_i(z)$ denote read and write operations respectively on a data item z by a transaction T_i . Consider the following two schedules.

- $S_1 : r_1(x)r_1(y)r_2(x)r_2(y)w_2(y)w_1(x)$
- $S_2 : r_1(x)r_2(x)r_2(y)w_2(y)r_1(y)w_1(x)$

Which one of the following options is correct?

- A. S_1 is conflict serializable, and S_2 is not conflict serializable
- B. S_1 is not conflict serializable, and S_2 is conflict serializable
- C. Both S_1 and S_2 are conflict serializable
- D. Neither S_1 nor S_2 is conflict serializable

Answer key

3.4.9 Conflict Serializable: GATE CSE 2021 | Set 2 | Question: 32

Let S be the following schedule of operations of three transactions T_1 , T_2 and T_3 in a relational database system:

$R_2(Y), R_1(X), R_3(Z), R_1(Y)W_1(X), R_2(Z), W_2(Y), R_3(X), W_3(Z)$

Consider the statements P and Q below:

- P : S is conflict-serializable.
- Q : If T_3 commits before T_1 finishes, then S is recoverable.

Which one of the following choices is correct?

- A. Both P and Q are true
- B. P is true and Q is false
- C. P is false and Q is true
- D. Both P and Q are false

Answer key

3.4.10 Conflict Serializable: GATE CSE 2022 | Question: 29

Let $R_i(z)$ and $W_i(z)$ denote read and write operations on a data element z by a transaction T_i , respectively. Consider the schedule S with four transactions.

$S : R_4(x)R_2(x)R_3(x)R_1(y)W_1(y)W_2(x)W_3(y)R_4(y)$

Which one of the following serial schedules is conflict equivalent to S ?

- A. $T_1 \rightarrow T_3 \rightarrow T_4 \rightarrow T_2$
- B. $T_1 \rightarrow T_4 \rightarrow T_3 \rightarrow T_2$
- C. $T_4 \rightarrow T_1 \rightarrow T_3 \rightarrow T_2$
- D. $T_3 \rightarrow T_1 \rightarrow T_4 \rightarrow T_2$

Answer key

3.4.11 Conflict Serializable: GATE CSE 2024 | Set 1 | Question: 36

Consider the following read-write schedule S over three transactions T_1, T_2 , and T_3 , where the subscripts in the schedule indicate transaction IDs:

$S : r_1(z); w_1(z); r_2(x); r_3(y); w_3(y); r_2(y); w_2(x); w_2(y);$

Which of the following transaction schedules is/are conflict equivalent to S ?

- A. $T_1T_2T_3$
- B. $T_1T_3T_2$
- C. $T_3T_2T_1$
- D. $T_3T_1T_2$

Answer key

3.4.12 Conflict Serializable: GATE CSE 2025 | Set 2 | Question: 43



Consider the database transactions T1 and T2 , and data items X and Y . Which of the schedule(s) is/are conflict serializable?

Transaction T1

R1(X)
W1(Y)
R1(X)
W1(X)
COMMIT(T1)

Transaction T2

W2(X)
W2(Y)
COMMIT(T2)

2

- A. R1(X), W2(X), W1(Y), W2(Y), R1(X), W1(X), COMMIT(T2), COMMIT(T1) 3
 B. W2(X), R1(X), W2(Y), W1(Y), R1(X), COMMIT(T2), W1(X), COMMIT(T1)
 C. R1(X), W1(Y), W2(X), W2(Y), R1(X), W1(X), COMMIT(T1), COMMIT(T2)
 D. W2(X), R1(X), W1(Y), W2(Y), R1(X), COMMIT(T2), W1(X), COMMIT(T1)

Answer key

3.5

Database Design (1)

3.5.1 Database Design: GATE CSE 1994 | Question: 3.11



State True or False with reason 4

Logical data independence is easier to achieve than physical data independence. 5

Answer key

3.6

Database Normalization (56)

Practice Tests: Test 1 (15Q) Test 2 (15Q) Test 3 (15Q) Test 4 (6Q)

3.6.1 Database Normalization: GATE CSE 1987 | Question: 2n



State whether the following statements are TRUE or FALSE: 7

A relation r with schema (X, Y) satisfies the function dependency $X \rightarrow Y$, The tuples $\langle 1, 2 \rangle$ and $\langle 2, 2 \rangle$ can both be in r simultaneously. 8

Answer key

3.6.2 Database Normalization: GATE CSE 1988 | Question: 12i



What are the three axioms of functional dependency for the relational databases given by Armstrong.

Answer key

3.6.3 Database Normalization: GATE CSE 1988 | Question: 12iia



Using Armstrong's axioms of functional dependency derive the following rules:

$$\{x \rightarrow y, x \rightarrow z\} \models x \rightarrow yz$$

1

(Note: $x \rightarrow y$ denotes y is functionally dependent on x , $z \subseteq y$ denotes z is subset of y , and $\models$ means derives).

gate1988 easy descriptive databases database-normalization 2

Answer key

3.6.4 Database Normalization: GATE CSE 1988 | Question: 12iib

Using Armstrong's axioms of functional dependency derive the following rules: 3

$$\{x \rightarrow y, wy \rightarrow z\} \models xw \rightarrow z$$

4

(Note: $x \rightarrow y$ denotes y is functionally dependent on x , $z \subseteq y$ denotes z is subset of y , and $\models$ means derives). 5

gate1988 normal descriptive databases database-normalization 6

Answer key

3.6.5 Database Normalization: GATE CSE 1988 | Question: 12iic

Using Armstrong's axioms of functional dependency derive the following rules: 7

$$\{x \rightarrow y, z \subseteq y\} \models x \rightarrow z$$

8

(Note: $x \rightarrow y$ denotes y is functionally dependent on x , $z \subseteq y$ denotes z is subset of y , and $\models$ means derives).

gate1988 normal descriptive databases database-normalization 9

Answer key

3.6.6 Database Normalization: GATE CSE 1990 | Question: 2-iv

Match the pairs in the following questions: 10

(a) Secondary index	(p) Function dependency
(b) Non-procedural query language	(q) B-tree
(c) Closure of a set of attributes	(r) Domain calculus
(d) Natural join	(s) Relational algebraic operations

11

gate1990 match-the-following database-normalization databases 12

Answer key 13

3.6.7 Database Normalization: GATE CSE 1990 | Question: 3-ii

Indicate which of the following statements are true: 14

A relational database which is in 3NF may still have undesirable data redundancy because there may exist: 15

- A. Transitive functional dependencies
- B. Non-trivial functional dependencies involving prime attributes on the right-side.
- C. Non-trivial functional dependencies involving prime attributes only on the left-side.
- D. Non-trivial functional dependencies involving only prime attributes.

16

gate1990 normal databases database-normalization multiple-selects 17

Answer key

3.6.8 Database Normalization: GATE CSE 1994 | Question: 3.6

State True or False with reason

There is always a decomposition into Boyce-Codd normal form (BCNF) that is lossless and dependency preserving.

Answer key

3.6.9 Database Normalization: GATE CSE 1995 | Question: 26

Consider the relation scheme $R(A, B, C)$ with the following functional dependencies: ¹

- $A, B \rightarrow C$, ²
- $C \rightarrow A$

- A. Show that the scheme R is in 3NF but not in BCNF. ³
 B. Determine the minimal keys of relation R .

Answer key

3.6.10 Database Normalization: GATE CSE 1997 | Question: 6.9

For a database relation $R(a, b, c, d)$, where the domains a, b, c, d include only atomic values, only the following functional dependencies and those that can be inferred from them hold ⁵

- $a \rightarrow c$ ⁶
- $b \rightarrow d$

This relation is ⁷

- A. in first normal form but not in second normal form B. in second normal form but not in first normal form ⁸
 C. in third normal form D. none of the above

Answer key

3.6.11 Database Normalization: GATE CSE 1998 | Question: 1.34

Which normal form is considered adequate for normal relational database design? ¹¹

- A. $2NF$ B. $5NF$ C. $4NF$ D. $3NF$ ¹²

Answer key

3.6.12 Database Normalization: GATE CSE 1998 | Question: 26

Consider the following database relations containing the attributes ¹⁴

- Book_id ¹⁵
- Subject_Category_of_book
- Name_of_Author
- Nationality_of_Author

With Book_id as the primary key. ¹⁶

- a. What is the highest normal form satisfied by this relation? ¹⁷
 b. Suppose the attributes Book_title and Author_address are added to the relation, and the primary key is changed to $\{Name_of_Author, Book_title\}$, what will be the highest normal form satisfied by the relation? ¹⁸

Answer key

3.6.13 Database Normalization: GATE CSE 1999 | Question: 1.24

Let $R = (A, B, C, D, E, F)$ be a relation scheme with the following dependencies

$C \rightarrow F, E \rightarrow A, EC \rightarrow D, A \rightarrow B$. Which one of the following is a key for R ? ¹

- A. CD B. EC C. AE D. AC ²

gate1999 databases database-normalization easy

Answer key

3.6.14 Database Normalization: GATE CSE 1999 | Question: 2.7, UGCNET-June2014-III: 25



Consider the schema $R = (S, T, U, V)$ and the dependencies $S \rightarrow T, T \rightarrow U, U \rightarrow V$ and $V \rightarrow S$. Let $R = (R1 \text{ and } R2)$ be a decomposition such that $R1 \cap R2 \neq \phi$. The decomposition is ⁴

- A. not in 2NF B. in 2NF but not 3NF ⁵
C. in 3NF but not in 2NF D. in both 2NF and 3NF

gate1999 databases database-normalization normal ugcnetjune2014iii ⁶

Answer key

3.6.15 Database Normalization: GATE CSE 2000 | Question: 2.24



Given the following relation instance. ⁷

X	Y	Z
1	4	2
1	5	3
1	6	3
3	2	2

⁸

Which of the following functional dependencies are satisfied by the instance? ⁹

- A. $XY \rightarrow Z$ and $Z \rightarrow Y$ B. $YZ \rightarrow X$ and $Y \rightarrow Z$ ¹⁰
C. $YZ \rightarrow X$ and $X \rightarrow Z$ D. $XZ \rightarrow Y$ and $Y \rightarrow X$

gatecse-2000 databases database-normalization easy ¹¹

Answer key

3.6.16 Database Normalization: GATE CSE 2001 | Question: 2.23



$R(A, B, C, D)$ is a relation. Which of the following does not have a lossless join, dependency preserving $BCNF$ decomposition? ¹²

- A. $A \rightarrow B, B \rightarrow CD$ B. $A \rightarrow B, B \rightarrow C, C \rightarrow D$ ¹³
C. $AB \rightarrow C, C \rightarrow AD$ D. $A \rightarrow BCD$

gatecse-2001 databases database-normalization normal ¹⁴

Answer key

3.6.17 Database Normalization: GATE CSE 2002 | Question: 1.19



Relation R with an associated set of functional dependencies, F , is decomposed into $BCNF$. The redundancy (arising out of functional dependencies) in the resulting set of relations is ¹⁵

- A. Zero ¹⁶
B. More than zero but less than that of an equivalent 3NF decomposition
C. Proportional to the size of F^+
D. Indeterminate

gatecse-2002 databases database-normalization normal ¹⁷

Answer key

3.6.18 Database Normalization: GATE CSE 2002 | Question: 16



For relation $R = (L, M, N, O, P)$, the following dependencies hold:

$M \rightarrow O, NO \rightarrow P, P \rightarrow L$ and $L \rightarrow MN$ 1
 R is decomposed into $R1 = (L, M, N, P)$ and $R2 = (M, O)$.

- 2
- Is the above decomposition a lossless-join decomposition? Explain.
 - Is the above decomposition dependency-preserving? If not, list all the dependencies that are not preserved.
 - What is the highest normal form satisfied by the above decomposition?

gatecse-2002 databases database-normalization normal descriptive 3

Answer key

3.6.19 Database Normalization: GATE CSE 2002 | Question: 2.24



5
 Relation R is decomposed using a set of functional dependencies, F , and relation S is decomposed using another set of functional dependencies, G . One decomposition is definitely BCNF, the other is definitely 3NF, but it is not known which is which. To make a guaranteed identification, which one of the following tests should be used on the decompositions? (Assume that the closures of F and G are available).

- Dependency-preservation
 - Lossless-join
 - BCNF definition
 - 3NF definition
- 6

gatecse-2002 databases database-normalization easy 7

Answer key

3.6.20 Database Normalization: GATE CSE 2002 | Question: 2.25



8
 From the following instance of a relation schema $R(A, B, C)$, we can conclude that:

9

A	B	C
1	1	1
1	1	0
2	3	2
2	3	2

- A functionally determines B and B functionally determines C
 - A functionally determines B and B does not functionally determine C
 - B does not functionally determine C
 - A does not functionally determine B and B does not functionally determine C
- 10

gatecse-2002 databases database-normalization

Answer key

3.6.21 Database Normalization: GATE CSE 2003 | Question: 85



11
 Consider the following functional dependencies in a database.

12

$Date_of_Birth \rightarrow Age$	$Age \rightarrow Eligibility$
$Name \rightarrow Roll_number$	$Roll_number \rightarrow Name$
$Course_number \rightarrow Course_name$	$Course_number \rightarrow Instructor$
$(Roll_number, Course_number) \rightarrow Grade$	

The relation $(Roll_number, Name, Date_of_birth, Age)$ is 13

- in second normal form but not in third normal form
 - in third normal form but not in BCNF
 - in BCNF
 - in none of the above
- 14

gatecse-2003 databases database-normalization normal

Answer key

3.6.22 Database Normalization: GATE CSE 2004 | Question: 50



The relation scheme Student Performance (name, courseNo, rollNo, grade) has the following functional dependencies:

- name, courseNo, $\rightarrow$ grade ²
- rollNo, courseNo $\rightarrow$ grade
- name $\rightarrow$ rollNo
- rollNo $\rightarrow$ name

The highest normal form of this relation scheme is ³

- A. 2NF B. 3NF C. BCNF D. 4NF ⁴

gatecse-2004 databases database-normalization normal ⁵

Answer key

3.6.23 Database Normalization: GATE CSE 2005 | Question: 29, UGCNET-June2015-III: 9



Which one of the following statements about normal forms is FALSE? ⁶

- A. BCNF is stricter than 3NF ⁷
- B. Lossless, dependency-preserving decomposition into 3NF is always possible
- C. Lossless, dependency-preserving decomposition into BCNF is always possible
- D. Any relation with two attributes is in BCNF

gatecse-2005 databases database-normalization easy ugcnetcse-june2015-paper3 ⁸

Answer key

3.6.24 Database Normalization: GATE CSE 2006 | Question: 70



The following functional dependencies are given: ⁹

$AB \rightarrow CD, AF \rightarrow D, DE \rightarrow F, C \rightarrow G, F \rightarrow E, G \rightarrow A$ ¹⁰

Which one of the following options is false? ¹¹

- A. $\{CF\}^* = \{ACDEFG\}$
- B. $\{BG\}^* = \{ABCDG\}$ ¹²
- C. $\{AF\}^* = \{ACDEFG\}$
- D. $\{AB\}^* = \{ABCDG\}$

gatecse-2006 databases database-normalization normal ¹³

Answer key ¹⁴

3.6.25 Database Normalization: GATE CSE 2007 | Question: 62, UGCNET-June2014-II: 47

¹⁵



Which one of the following statements is FALSE? ¹⁶

- A. Any relation with two attributes is in BCNF ¹⁷
- B. A relation in which every key has only one attribute is in 2NF
- C. A prime attribute can be transitively dependent on a key in a 3NF relation
- D. A prime attribute can be transitively dependent on a key in a BCNF relation

gatecse-2007 databases database-normalization normal ugcnetcse-june2014-paper2 ¹⁸

Answer key

3.6.26 Database Normalization: GATE CSE 2008 | Question: 69



Consider the following relational schemes for a library database:

Book (Title, Author, Catalog_no, Publisher, Year, Price)
Collection (Title, Author, Catalog_no)